CHAPTER 9

Monday: February 16th 2026

Introduction

Context

This lecture marks the beginning of **Analysis II: mehrere Variablen** (Multivariable Calculus) for the Spring 2026 semester at ETH Zürich. The lecturer, Joaquim Serra, introduces the course structure, logistics, and provides a high-level overview comparing the fundamental goals of Analysis I (single-variable calculus) with the upcoming objectives of Analysis II (vector calculus and higher-dimensional integration).

Introduction and Administrative Information

Spoken Protocol [00:00:00 - 00:01:20]

Good morning. Good morning. Thank you. Please, please, please. Uh, good morning and welcome to Analysis 2. Uh, I have here a projection with my name, Joaquim Serra, the name of the coordinator, Enric Florit-Simón. Now, this is my introduction. Um, and then you know the times of the lectures, obviously, because you are here. Um, and I wanted to share before starting with the mathematical content some practical informations that are very important because we are a large group and the — the communication and everything needs to be done in a more or less structured way. Otherwise, otherwise we collapse.

Projected Content

The lecturer displays the course website for *"Analysis II: mehrere Variablen Spring 2026"*. Key details shown:
Lecturer: Joaquim Serra
Coordinator: Enric Florit-Simón
Lectures: Mon 08:15–10:00 (ETA F 5), Wed 08:15–10:00 (HG F 7 / HG F 5), Thu 16:15–18:00 (ETA F 5).

Spoken Protocol [00:01:20 - 00:03:30]

So, as I guess you did — so as a general rule, uh, everything that applied to Analysis 1 applies to Analysis 2, unless otherwise stated. We will try to follow the same rules and then you know what to expect. In particular, you will have this, um, metadata page that I can see that from — from up there is a bit difficult to see, right? But I will try to put it bigger, yeah. Um, so you know that you need to register for the exercise classes. Uh, for the communication, please try to use whenever possible the forum. We will look at the forum and — and reply, and also your — your peers can also reply to questions because many — so then we can use the collective knowledge, right? Um, there will be recordings of the lecture, but not until the later on, okay? Because, uh, is important for the lecture and for you, but mainly, I mean, for you is your interest, uh, if you want to come or not. But for the lecture is very good that you come to the lecture, you ask questions, you stop me when I'm going too fast, these kind of things. You know, it's very important. So it's a — and also is where you meet with your peers and you can communicate, ask questions to your peers. It's — it's — it's better for the learning.

Spoken Protocol [00:03:30 - 00:05:07]

Um, we will tell you more about the exam later on, right? Today I will tell you a bit more what is important to be successful in the course, what to expect about this course, and, uh, the details and the rules of the exam will come later on. You don't need to know now. The bonus, there is a 0.25 bonus, will be in questions that you will answer in some exercise classes. We will tell you in advance which ones. And, uh, essentially you need to be in the — in the exercise class and answer it. And, uh, do not worry because, uh, the — the rule will be thought — in any case it's just 0.25 points — uh, but the rules will be thought so that if you miss one day or casuistics like this, it's not that you get zero points, right? Okay. Um, lecture notes. There will be lecture notes similar to those in Analysis 1, and we will put them in this page, hopefully by the end of each week, because I — I update them after the lecture, mainly depending on your questions, depending on your feedback. And that's it, uh, I think that the rules are mainly those and the — the organization. Is there some question, uh, about organization? So essentially, as I said, uh, if you have doubts, uh, is like in Analysis 1, unless I say something that is surprising to you, as in Analysis 1. Is it okay, this? Yeah. Okay, now I will switch — I have some other things I want to project later, but now, because we cannot record at the same time blackboard and projection, I will switch one moment to the blackboard. *[The lecturer turns off the projector and prepares the blackboard]* So I can mute, I think. Yeah. Okay. Now we need the light. *[Adjusts classroom lighting]* Okay. Ah, I think that there is already on. Okay.

Overview: Analysis I vs. Analysis II

Spoken Protocol [00:05:07 - 00:07:20]

Okay, now, just to start, um, we will give a very quick, so quick that is almost a cartoon, but a very quick overview of what is Analysis 1 and Analysis 2, right? It's very, very quick. *[Starts writing on the left blackboard]* So, Analysis 1. Um, is — sorry, but for me is very difficult to write big and then it's — so — so if — if you don't see something, I will try to say at loud everything I write, but if — if something — I mean, Analysis is... so, um, Analysis 1, calculus, sorry, it's the first day, right? Bear with me. Calculus is written

— is spelled correctly? Yeah. Calculus of one variable, right? And then, if I ne — if — if we had to choose a representative theorem of Analysis 1 that a bit represents the course, uh, what would you choose? So, can you tell me — I know it's the first day, but let's start a bit of communication. Can you tell me at least two answers — so two people can tell me what is the representative — one representative theorem of Analysis 1? Yeah?

Student Interaction

Fundamental Theorem of Calculus.

Spoken Protocol [continued]

Yeah. I agree. Some other representative theorem? He said the Fundamental Theorem of Calculus. Makes sense, right? Okay, after — is there some other guess? Is — yeah, there?

Student Answer

The Intermediate Value Theorem.

Spoken Protocol [continued]

The Intermediate Value Theorem. Okay. I think we need to use this for the recording, otherwise I — I will repeat. [*Referring to the catch-box microphone*] Later on when I get — I get better with the throwing the — we will use that. So, okay, so fundamental — so we said two theorems that are — that are indeed important. Perhaps the most representative, I would agree and — and I saw many people saying yes, so is the Fundamental Theorem of Calculus, right? And is this... [*Starts writing the formula on the board*]

Board Action: The Fundamental Theorem of Calculus

The lecturer writes the Fundamental Theorem of Calculus (FTC) on the board. He initially makes a common sign error in the subtraction, which is subsequently corrected.

The Fundamental Theorem of Calculus (Recall)

In Analysis I, the central result is the Fundamental Theorem of Calculus (FTC), which relates the integral of a derivative to the values of the function at the boundaries:

$$\text{FTC} \qquad \int_a^b f'(x)\,dx = f(a) - f(b) \qquad \text{(Incorrect)}$$

Spoken Protocol [00:08:35 - 00:08:50]

Is it correct? Yeah, looks like... [*Pauses and looks at the board*] Yeah, there is some mistake? [*Corrects the board*] Good, well spotted.

The Fundamental Theorem of Calculus (Corrected)

The corrected statement of the theorem:

$$\text{FTC} \qquad \int_a^b f'(x)\,dx = f(b) - f(a)$$

Spoken Protocol [00:08:50 - 00:10:32]

Now, to — to unfold and to write and to understand every detail about this equation over here, you needed to introduce a lot of things. You needed to define what is that (i.e., the derivative f'), right? What is this (i.e., the Riemann integral $\int_a^b$)? And even — and you spent a lot of time, right? — what is that (i.e., the real numbers $\mathbb{R}$)? It was not — it was not easy. Yeah. Okay, so this is a fairly, fairly compact theorem, yet the ingredients you had to learn, uh, to unfold all of this and to understand the meaning of every symbol here were a bit of work. Now, how will Analysis 2 look like? *[Moves to the next section of the board]* So this will be calculus, calculus, still calculus in, uh, more than one variable. Well, since you already know one, let's say at least two variables, because otherwise is already covered by Analysis 1. And some representative theorem that would be the equivalent — so the counterpart of this one (i.e., the FTC) — is the Gauss — it has many names, and is always the same theorem. So but — so is the Gauss, Stokes, uh, what else? Divergence, sometimes is called the Divergence Theorem. Uh, and — and that's it, I think. Green, sometimes Green Theorem.

Analysis II: The Goal

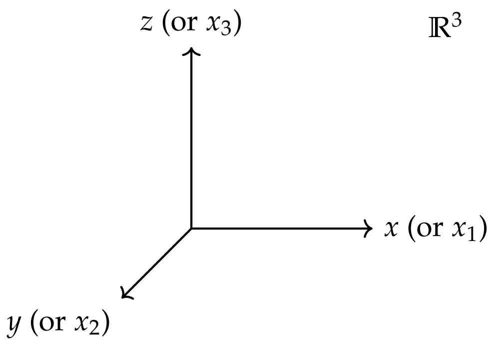
Analysis II extends these concepts to higher dimensions:
Analysis 1: Calculus of one variable.
Analysis 2: Calculus in ≥ 2 variables.
The representative results in Analysis II are a family of related theorems:

- Gauss Theorem
- Stokes Theorem
- Divergence Theorem
- Green Theorem

Spoken Protocol [00:10:32 - 00:11:00]

And now I will write the theorem, and you will not understand exactly what each thing means. I will tell you intuitively, and then the goal of the lecture will be at the end of the lecture you will be able to understand in full precision everything that I wrote here. Okay? So the theorem goes as follows. So we have — we have the space, the space, let's say I will put it in the space $\mathbb{R}^3$. Three coordinates of space, right? This thing... *[Draws a 3D coordinate system on the board]*

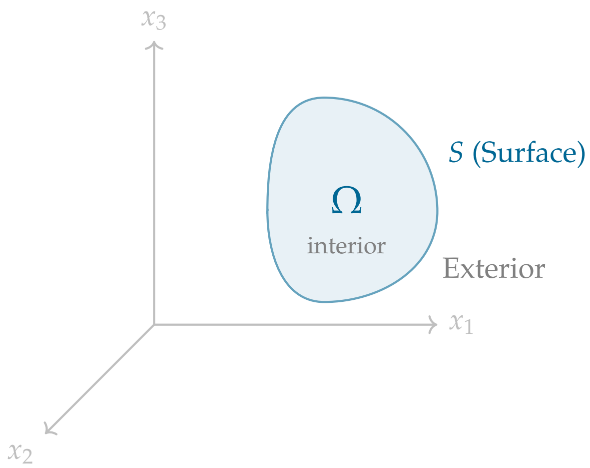
Visualizing the Higher-Dimensional Space



Spoken Protocol [00:11:00 - 00:12:28]

Right? And then in the — in this space (i.e., $\mathbb{R}^3$) we are given a domain, we are given some — or some surface that encloses some interior — so this — this is the interior of the surface and the exterior. Exterior. And this is a surface, right? The surface I call it S. What is a surface? It could be a sphere, it could be a sort of — the skin of a *"potato"*. It's a surface. Two-dimensional object. Right? It has and — and what I'm — this kind of surface will have a interior and a exterior. And the interior of the surface — so the — if the surface is the skin of the potato, the filling of the potato, the white part, right? Is Ω. I call it Ω.

Visualizing the Domain and Surface



Explanation of Steps: The lecturer uses the *"potato"* analogy to describe a bounded volume $\Omega \subset \mathbb{R}^3$ enclosed by a two-dimensional surface $S = \partial\Omega$. This geometric setup is foundational for the higher-dimensional versions of the Fundamental Theorem of Calculus.

Spoken Protocol [00:12:28 - 00:13:32]

And then the equivalent of this theorem (i.e., the FTC) in this setup is this. Given a vector field — now you don't know what it is a vector field, let me tell you. f equal f_1, f_2, f_3. These are three — three functions of the space, right? At each point of space you have three numbers. Here you have three numbers, here you have three numbers.

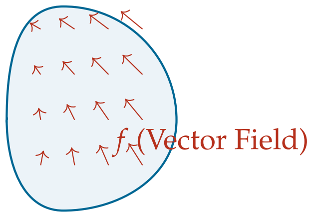
Yeah. This is a vector field and how we — do we draw vector fields? Um, we draw it like this, right? It's like a — a field of some cereal, right? But the — in every point you have an arrow instead of a — a plant.

Defining the Vector Field

A vector field f assigns a vector to every point in space:

$$f = (f_1, f_2, f_3)$$

where each $f_i : \mathbb{R}^3 \to \mathbb{R}$ is a scalar function.



Spoken Protocol [00:13:32 - 00:15:15]

Okay, so this is the vector field f. And the theorem says that the integral in Ω of the — the derivative of the field, the i component with respect to the x_i variable, and you sum over i. This is equal to the flux of the field across the boundary, across the S. And it's called like this. Is $f \cdot \nu$ differential of surface. This is the same theorem you have got here in one dimension (Recall: FTC) in several variables.

The Divergence Theorem (Gauss's Theorem)

The *"final boss"* of Analysis II relates the volume integral of the divergence to the surface integral of the flux:

Divergence Theorem

$$\int_\Omega \left(\sum_i \frac{\partial f_i}{\partial x_i} \right) dx = \int_S f \cdot \nu \, dS$$

Explanation of Steps: This equation is the higher-dimensional analogue of the Fundamental Theorem of Calculus. It states that the total *"source"* or *"sink"* of a vector field within a volume Ω (the left side) is exactly equal to the net flow or *"flux"* of that field through the enclosing surface S (the right side).

Spoken Protocol [00:15:15 - 00:17:38]

Now the small problem is that maybe some of you in physics or — or in previous courses have seen some of these notations, probably not. I don't expect you to — we — and we will need to introduce each of these notations. These partial derivatives, what is a domain, what is a surface? What is this surface mathematically? How do you model a surface? How do you compute a integral over a surface? You know how to compute a integral over the real line, right? Over a segment, but over a surface? This will need some definition and some work. And these objects that are the partial derivatives. So here in — in one equation you have a bit the goal and the syllabus of the course. How will — how will we proceed? So first, we will start looking at the space. In Analysis 1, you did everything in $\mathbb{R}$. Now we will need to introduce $\mathbb{R}^3$, but we are mathematicians so we will — well, some of us. Some of you are physicists or are proto-physicists, would like to become physicists. So we will use this because we don't want to spend time proving the same theorem in — in the plane $\mathbb{R}^2$, then in $\mathbb{R}^3$ in the space, and then when you go to do general relativity you need $\mathbb{R}^4$, right? We do everything in $\mathbb{R}^n$. n is the dimension. So what is $\mathbb{R}^n$? Is the set of n — n-tuples of real numbers. This is a definition. Okay? And what is — this is the same as the set of points — we will denote a point x, but now x will carry n coordinates. $x_1, x_2, \dots, x_n$. Now this is a point. And these x_i are real numbers. This you know probably because you — you are doing or you have done linear algebra, right? So this should be quite familiar. But still is good that we recall.

Definition of $\mathbb{R}^n$

Definition (The Space $\mathbb{R}^n$)**.** The space $\mathbb{R}^n$ is defined as the set of all n-tuples of real numbers:

$$\mathbb{R}^n = \{n\text{-tuples of real numbers}\}$$

Formally, a point $x \in \mathbb{R}^n$ is represented by its coordinates:

$$x = (x_1, x_2, \dots, x_n) \quad \text{where } x_i \in \mathbb{R} \text{ for all } i \in \{1, \dots, n\}$$

Spoken Protocol [00:17:38 - 00:21:13]

Okay. Now, the first part of the course we will be studying $\mathbb{R}^n$ and the properties of $\mathbb{R}^n$. The same a bit as in $\mathbb{R}$ you studied the — the axiom of the supremum, right? Or the — the fact that you can take, uh, limits of Cauchy sequences, these kind of facts, right? The same we will need to study in $\mathbb{R}^n$. Then there will be some new things or — or things that you could also study in $\mathbb{R}$ but become very trivial and now will be much more interesting is the topology of $\mathbb{R}^n$. So first, the $\mathbb{R}^n$ as a space, as Euclidean space. So this is a bit the syllabus. 2: the topology of $\mathbb{R}^n$. You will see what this means indeed, the topology. Then 3: this object over here *[points at the partial derivatives in the Divergence Theorem]* (i.e., the partial derivatives). Partial derivatives. So when you have — when you have several variables you can differentiate a quantity with respect to each of them, right? And differential and — so the differentiation of functions of several variables. Then 4: integration. Integration in $\mathbb{R}^n$, here in a domain of $\mathbb{R}^n$, and on surfaces. But because we want to learn also how to integrate on surfaces, we will do a bit of introduction to differential geometry. Introduction to, uh, surfaces and *"submanifolds"*. Now I know it's a lot of words, but we will introduce one by one. So surface is a usual two-dimensional object, right? Typically smooth. When it is a three-dimensional object in $\mathbb{R}^4$, for instance, we call this a submanifold. You can also call it surface but — but it's more precise the word submanifold. Or — or — or some two-dimensional object in a five-dimensional space. The — any — any combination of dimensions. And then when you know this, when you know, uh, yes, you will need to know how to integrate. Integration over $\mathbb{R}^n$ and surfaces.

Course Syllabus

The roadmap for Analysis II consists of five major pillars:

(a) $\mathbb{R}^n$ as Euclidean space.

(b) Topology of $\mathbb{R}^n$.

(c) Partial derivatives and differentiation of multivariable functions.

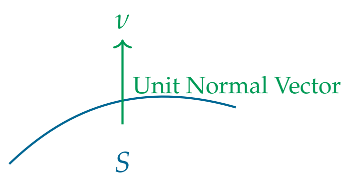
(d) Introduction to surfaces and *"submanifolds"* (Differential Geometry).

(e) Integration over $\mathbb{R}^n$ and surfaces.

Spoken Protocol [00:21:13 - 00:22:00]

And hopefully then you will know all of these elements. This element, uh, this is the normal vector. This I can tell you now what is this. This is just the vector, the unit vector, the vector of length one that is normal to the surface, right? If you have a surface, this — this ν is just the — the normal to the surface. So this is easy. Um, but still we will — we will define this in detail. And after we know all of these, we will be able to prove that equation over here (i.e., the Divergence Theorem) and work with it and see why it is useful. Right? For many things. Okay. Uh, so this is a rough plan of the course.

The Normal Vector



The Role of Proofs in Analysis

Spoken Protocol [00:22:00 - 00:24:15]

Now I wanted to — to do a very small survey about your attitude, um, attitude towards proofs. Okay? This is important to mention that. *[Moves to the right chalkboard]* So I will put two columns. Column left is *"I find proofs"* — so proofs I mean rigorous, long, very careful mathematical proofs with all details. Okay? This is mean proofs. So here I will put: I find proof, um, essential, illuminating, useful to clarify, the best part of any course. And on this column over here I will put: I find the proof innecessary, confusing, mysterious, I don't know when I have finished the proof or not, right? Because it looks like — it sounds convincing but — mysterious, no clear — no clear rules of the game.

Survey: Attitude Towards Proofs

The lecturer creates a comparative table on the board to gauge student sentiment regarding formal mathematical proofs.

Positive / Essential	Negative / Challenging
Essential	Innecessary
Illuminating	Confusing
Useful to clarify	Mysterious
The best part	No clear rules of the game

Spoken Protocol [00:24:15 - 00:26:15]

And now, okay, this is extreme — I mean, I'm — I'm making all this on purpose. This is extreme, right? This is extreme. I know that any — most of you would be maybe in the middle. But I would like to know which of you lean more towards this opinion or more towards this opinion. Okay? So let's start with this one. So who is here? Who loves proofs and... okay. *[Students raise hands]* Okay. And here... *[Students raise hands for the other column]* Okay, it's interesting. So it's roughly — there are a bit more people that love proofs, but — but okay. So there is no correct answer, right? Uh, so I would be — well, depending on where — in between.

Spoken Protocol [00:26:15 - 00:28:30]

So why — why am I asking this? So does it coin — is it — does it have some correlation with physicists and mathematicians or not? Sometimes. What do you think? It — it could make sense, not always. There are physicists that... but — it could make sense. So I think that, um, about this discussion of proofs, I wanted to — to use ten minutes today to tell you two things. First, why I believe — many people can have many answers — but why I believe that the proofs are important and what will we test in this course, right? Because it's — it's a bit correlated of course.

So the first thing is that in — in algebra many times you have a theorem that is very beautiful, very clean, and you can apply in many situations without modifying it, right? In analysis, we will see that, sometimes the theorem that you learned doesn't exactly apply to the situation you — you need. Then you need to know how to modify a little bit the theorem. And the only way to do that is to know the proof and then modify a — some step here and some step there.

***Insight:* Analysis as a Malleable Tool**

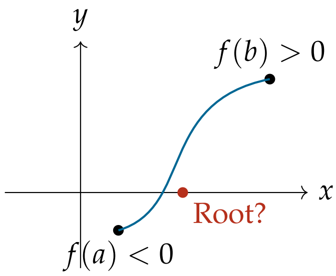
The lecturer highlights a fundamental difference between Algebra and Analysis: in Analysis, theorems are often not *"black boxes"* but templates. Understanding the proof is what grants the mathematician the flexibility to adapt a result to non-standard domains or boundary conditions.

Spoken Protocol [00:28:30 - 00:31:17]

Also, many times the proof contains the ideas on why the theorem is true, right? The theorem is a bunch of symbols as we wrote there sometimes or statements, but it's not very illuminating. It's okay to know that a fact is true, but when you know a bit why, it's — it's — it's more useful many times, right? Being able to export and to use in situations that are new and also, um, understanding better, meaning being able also sometimes to

remember the — the theorem. Because if you — if you get quick at remembering proofs and arguments, sometimes you don't remember all the assumptions of a theorem and you can figure — you can figure them out just by — by repeating the proof in — in your mind, right?
This doesn't mean that you need to learn the proof in all the details all the time. It's very important — so that the proof is useful and illuminating and so on — it's very important that when you have this super long proof, you extract the main ideas of the proof. A very short — so you need to have several levels in your head. So a proof that is that long, you can remember three sentences like hints that are the key ingredients of the proof for you. And it's very good exercise to do that. Then you expand. You learn how to expand these — these summaries. Is it clear that? And how you learn how to expand? You learn how to expand, um, hints by doing exercises. That's why the exercises are important. If you become very good in solving exercises, you'll be able to compress the information very well because then you don't need to remember all the details. You just remember the highlights of the proof and then you fill the details because you are expert in doing that, right? So becoming expert in exercises is — is good to find the proof also more interesting and more useful and not that, um, much of an effort.
Now, some people that are here that find the proof confusing, mysterious, innecessary, is because sometimes the — the rules of the game and why the game is interesting is not that clear. So why when you finish a proof, why can you say that the proof is finished, right? Because the professor says so that this the proof QED? Is common sense? Um, what is — most people see that, oh, this looks like a argument that is conclusive. Most people. But not every time. And there are very subtle arguments. So that's why we need to spend sometimes a lot of time with things that seem obvious because I tell you: a continuous function crosses the zero axis at some point (Recall: Intermediate Value Theorem). Okay, of course it crosses. But — this intermediate value theorem, right? Continuous function crosses the — of course, but you need to prove it, right? And the proof cannot be just the intuition of I make a drawing... *[Draws a line crossing an axis]* I have the zero line, I have two points, one here, one here, I have a continuous function, so if I cannot hold the — the pen from the paper, I need to cross the line at some point. This is not a mathematical proof. But it's a correct intuition.

Intuition vs. Rigor



Explanation of Steps: The lecturer uses the Intermediate Value Theorem to illustrate that while geometric intuition (the *"pen on paper"* argument) is powerful, it does not constitute a formal proof. Rigorous analysis requires proving these *"obvious"* facts from foundational axioms.

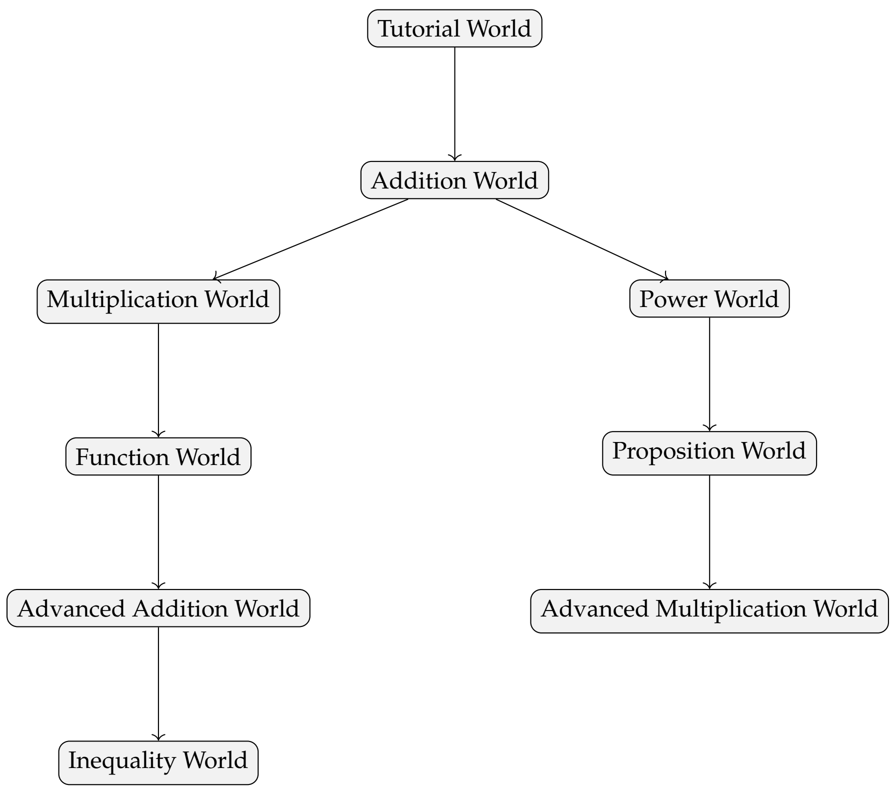
Spoken Protocol [00:31:17 - 00:31:17]

Now, so what can help with this? I think that something that can help with this, with the rules of what is a proof and what is not a proof... *[Turns on the projector]* Now we will need to turn on the projector again. Okay. Let me see. *[Projector starts up]* Hmm. Okay. So if someone is interested, you can go — you can Google, let's see, Google... you can Google, uh, Natural Number Game. *[Lecturer searches for the game on Google]* This is a serious stuff. I mean, I'm — I'm not — this is a very serious stuff. It's difficult. I mean, I know that there are some cartoons, but this is — is a serious difficult game. So — so it's — so this is a — so one — one argument and — and now seriously, in the light of artificial intelligence and — and all what is going on. So one argument that is very interesting about proofs is that — maybe some of you didn't realize — but a proof is computable. So if a proof is correct or not can be computed. That's why it's not an opinion, right? You can — there are several computation languages for... *[Audio cuts off abruptly]*

Projected Content: The Natural Number Game

The lecturer shows the *"Natural Number Game"* website, an interactive tool for learning formal proof verification using the Lean proof assistant. The interface shows a progression of *"worlds"* representing different mathematical concepts.

The Structure of Formal Proofs (Lean)



Spoken Protocol [00:00:00 - 00:01:56]

...people that find the proof confusing, mysterious, innecessary, is because sometimes the — the rules of the game and why the game is interesting is not that clear. So, why when you finish a proof, why can you say that the proof is finished, right? Because the professor says so that this is the proof Q.E.D.? Is common sense? Um, what is — most people see that, oh, this looks like a argument that is conclusive. Most people. But not every time. And there are very subtle arguments. So, that's why we need to spend sometimes a lot of time with things that seem obvious because I tell you: a continuous function crosses the zero axis at some point (Recall: Intermediate Value Theorem). Okay, of course it crosses. But — this intermediate value theorem, right? Continuous function crosses the — of course, but you need to prove it, right? And the proof cannot be just the intuition of I make a drawing — I have the zero line, I have two points, one here, one here, I have a continuous function, so if I cannot hold the — the pen from the paper, I need to cross the line at some point. This is not a mathematical proof. But it's a correct intuition. Now, so what can help with this? I think that something that can help with this, with the rules of what is a proof and what is not a proof... *[The lecturer turns on the projector and searches on Google]* Now we will need to turn on the projector again. Okay. Let me see. Um. Okay. So, if someone is interested, you can go — you can Google, let's see, Google... you can Google, uh, Natural Number Game. *[The lecturer navigates to the Natural Number Game website]*

Projected Content: Lean Game Server

The lecturer displays the *"Lean Game Server"* website, specifically the *"Natural Number Game"*. The site is described as a repository of learning games for the proof assistant Lean (Lean 4) and the mathematical library *"mathlib"*. Other games shown include *"Relativity"*, *"Field Theory"*, *"Thermodynamics"*, and *"Real Analysis, The Game"*.

Spoken Protocol [00:01:56 - 00:04:08]

So, if someone is interested, you can go — you can Google, let's see, Google... you can Google, uh, Natural Number Game. Is — this is a serious stuff. I mean, I'm — I'm not — this is a very serious stuff. It's difficult. I mean, I know that there are some cartoons, but this is — is a serious difficult game. So — so it's — so this is a — so one — one argument and — and now seriously, in the light of artificial intelligence and — and all what is going on. So, one argument that is very interesting about proofs is that — maybe some of you didn't realize — but a proof is computable. So, if a proof is correct or not can be computed. That's why it's not an opinion, right? You can — there are several computation languages for proofs, one of the most famous is this Lean, where you can write the — the proof in the language. It's very tedious to write the proof in the language, but now there are tools that help you. And now the — it's like a computer program. If the program compiles, the proof is correct up to the axioms. You provide the axioms, you provide the reasoning, and if the program compiles, it's — proof is correct. And if it doesn't compile, it doesn't mean that it's not correct. It means that you need to put more details. Maybe it's not correct, maybe yes, but it's not detailed enough. Right? So, first argument is that, um, so — or some important argument is that proof are computable, this means that this ends all the discussion. If I wrote a proof, it's true, that's it. End of the discussion. Okay? Which is interesting. So, it — it doesn't matter if Donald Trump says that the proof — the fundamental theorem — I mean, no, he can say whatever he wants, but the theorem will be true, okay? That nowadays is — this is an interesting aspect.

***Insight:* AI and the Verifiability of Proofs**

The lecturer uses the emergence of proof assistants like Lean to ground the concept of mathematical truth. By framing a proof as a *"computable"* object rather than a matter of opinion, he highlights the absolute rigor required in Analysis. This serves as a meta-commentary on the reliability of AI models (like ChatGPT or Gemini), which may *"hallucinate"* plausible-sounding but logically flawed arguments, whereas a compiled Lean proof is objectively verified.

Spoken Protocol [00:04:08 - 00:08:20]

This also an interesting aspect for these guys, right? *[Points at the ChatGPT and Gemini tabs on his browser]* Right? So, I — I had on purpose the — the ChatGPT or the Gemini or whatever. So, you are using those, right? You are using these — I know that some people tell you *"don't use that"* — well, I know you are using those. Um, and I think it's — I think you are — you — you know much better how to use than — than people think. Um, so — but you see, these guys still, especially with more advanced proofs, hallucinate. And say things sometimes that are wrong. And then knowing how to spot this is another motivation. Knowing how to spot wrong arguments and — and being careful with correct and incorrect arguments — um, is a good motivation to learn how to do correct proofs. Okay? And you can train by doing exercises, and if you are not sure, if you are — now I — I put it down. But if you are not sure what are the rules of the game, what is a proof, what is not a proof, try to — I know it's difficult, try to go to the Natural Number Game and play a bit, and you will see what is a fully rigorous proof up to — up to the end. And why we don't do this all the time. Because now — the next thing I wanted to tell you is in Analysis 2, the pace will be quicker than in Analysis 1. We'll start roughly at the same pace, do — doing many details, and then we will go a bit faster. What does it mean faster? That I will not be as detailed. You see? Some things is *"because this and that, then this follows"*. And then you — if you don't see that this follows, if many don't see, you ask question: *"Wow, uh, why this follows? I don't see — we don't see"*. Then we can provide details. But we will not be able to stop all the time to provide details because some of the arguments you already know how to do them. The Analysis 1 arguments, I will say *"okay, this is like in Analysis 1"*, right? So, the pace is quicker and quicker. And the more you advance in your studies, it will be quicker. This doesn't mean that you can be okay with not — uh, understanding, right? You need to — if some statement you don't know how to prove, you can try by yourself, you can ask your peers, you can unfold everything. And this is a very good exercise. So, you need to do your exercises, you need to try to understand all the steps of the proof, meaning that ideally you should be able with time and — and effort to — to go back to the axioms, right? You don't need to do every time. You — you just need to — to be — to know that you are able to. Okay? And then you are okay. So, if you can do this with — with most — proofs, um, it — it means that you are following very well the course. Okay?

Spoken Protocol [00:08:20 - 00:10:30]

Is there something else I wanted to tell you before we start with the lecture? I don't think so. So, we have roughly ten minutes before the pause. I will start with the proper — lecture — the second half. And then — uh, let me ask you, do you have some questions about what I said in this first part? A bit the plan of the course, how to be successful in the course, what do we expect from you? I — now I have the impression that I said many things and maybe — uh, I will need to insist — uh, later on and come back to the objectives. Let me summarize if — if there are no questions let me summarize in three points what are the — a bit the main learning goals. Okay? So, maybe point number one that is very relevant to both physics and — physicists and mathematicians is being able to do computations. Computations. Computations means — um, advanced computations. Compute the — the solution of some O.D.E., right? Several variable stuff. A computation can be compute some integral. Why the — the area of the sphere is — whatever, 4π or of the radius one, $4\pi r^2$? You need to be able to compute that integrals and — with a numeric — so with a — um, — with a concrete output like a numeric value. This — this could be a concrete goal. Then you need to be able to follow proofs. Follow the — the logic. And this you train with exercises, that some of them are computation — also the computations, but some of them are more computation, some of them are more theoretical. And also you can train with the lecture notes. Whenever you don't understand a passage of the lecture notes, as I said before, you train — train to fill out details. If you cannot, no problem, you ask, you ask your peers, then you ask me, and is a good exercise. Right? So, you have plenty of material to — of material to train.

Spoken Protocol [00:10:30 - 00:11:36]

And essentially these are the two — the two main legs: so computation and proofs. And then the third leg that I will introduce a bit is — a bit assessing if a proof is correct. This is something we didn't test in the past, but now it's becoming more and more important. Okay? So, some type of exercise that I will try to introduce at some point is giving you some proof with a flaw argument. Sometimes I make mistakes, so then it's good because you can practice this and then you see... but I will try to make less mistakes. And then if I would then do some proofs with incorrect arguments on purpose and then you can spot *"ah, this subtlety you — we didn't get it right"* or here — here the ChatGPT made a hallucination, okay? It's getting more and more harder and harder to get these models to hallucinate, but still in Analysis 2 it's — is harder now. But — but then we will put trickier questions. Um. Okay. So, it's five minutes. Do you have any question, someone? Ah, there? Is there a question? Yeah. Okay. Thank you. Sorry, I cannot see very well so...

Rigor in Tests

I wanted to ask how rigorous do the proofs have to be in the test? Do we have to do them like the Analysis 1 test where we have — we don't have to write down the axioms every time, or we just mention the most important theorem, or do we have to do very rigorous proofs?

Spoken Protocol [continued]

Let me — let me put the question about — about — let me put the question a bit different. The proof are always very rigorous. The only thing that they don't need to be as detailed as. You see? You can do some claims and you don't need to justify to the smallest detail every single claim. In Analysis 1, you see the difference of the theorem, it's clear.

Blackboard Transition

The lecturer erases the left and center chalkboards to summarize the transition from Analysis I to Analysis II.

Spoken Protocol [00:12:33 - 00:13:17]

So, what would be a — a proof of this? So, how would you summarize in a sentence a proof of — of this equation over here? Let's try this. *[Points at the FTC on the board]* Yeah. Fundamental Theorem of Calculus. *[Starts writing on the board]*

The Fundamental Theorem of Calculus vs. Divergence Theorem

The lecturer restates the central comparison between the two courses:

- **Analysis 1:** Calculus of one variable.

$$\text{FTC} \qquad \int_a^b f'(x)\,dx = f(b) - f(a)$$

- **Analysis 2:** Calculus in ≥ 2 variables.

$$\text{Gauss/Divergence} \qquad \int_\Omega \sum_{i=1}^n \frac{\partial f_i}{\partial x_i}\,dx = \int_S f \cdot \nu\,dS$$

Spoken Protocol [00:13:17 - 00:15:18]

So, anyone — someone want to tell me one sentence that kind of explains the idea of the proof of that? *[Points at the FTC]*

Student Answer

We can show that the derivative of the primitive converges to the actual value of the function at the — I think it was a or no, b.

Spoken Protocol [continued]

The derivative of the — yeah. So, the derivative with respect to this — this value, right? Yeah, this is a good — a good answer. So, if — so everyone has — this is what I was telling you about before. So, when you see a — something like this, the first time you see, you see all the proof, and then there — there needs to be something that you remember. That is some summary of the proof that for each person might be a bit different. There are also several ways to do the proof. And then with this, you are able to reconstruct the details. You want to know what — what I remember when I see this? I see — I always remember the *"telescopic sum"*. Do you know what is a telescopic sum, right? So, when I see this, I see — I see that. Right? I see... let me put it there.

***Insight:* The Telescopic Sum Analogy**

The lecturer provides a powerful mental anchor for the Fundamental Theorem of Calculus: the telescopic sum. By viewing the integral as a sum of infinitesimal increments where internal terms cancel out, leaving only the boundary values, the student gains a structural intuition that carries over to the Divergence Theorem in higher dimensions.

Intuition: The Telescopic Sum

The proof of the FTC can be visualized by partitioning the interval $[a,b]$ into N small sub-intervals:

$a = x_0$ $\qquad$ x_i $\qquad$ $b = x_N$

The difference $f(b) - f(a)$ is expanded as a sum of increments:

$$\begin{aligned} f(b) - f(a) &= f(x_N) - f(x_0) \\ &= \underbrace{(f(x_N) - f(x_{N-1})) + (f(x_{N-1}) - f(x_{N-2})) + \cdots + (f(x_1) - f(x_0))}_{\text{Telescopic Sum}} \\ &= \sum_{i=1}^{N} (f(x_i) - f(x_{i-1})) \end{aligned}$$

Explanation of Steps: By multiplying and dividing each term by the step size $h = x_i - x_{i-1}$, each increment $\frac{f(x_i)-f(x_{i-1})}{h}$ approximates the derivative $f'(x_i)$. In the limit as $h \to 0$, this sum becomes the Riemann integral $\int_a^b f'(x)\,dx$.

Spoken Protocol [00:15:18 - 00:17:15]

So, you are unfolding this sum. Now, if you call the distance between these two consecutive points h, you multiply and divide by h, right? And each of these pieces is approximately the derivative. End of the proof. Okay? I will not go at this speed. But this is a way to remember the proof. And you need to find your own way in each proof. And then it's not that hard to remember them. Okay, so it's nine. We do the 15 minutes pose and then we start with the proper lecture. *[The lecturer takes a break. The video resumes with the lecturer at the board]* So, I will — I will do my best, I will do my best to respect the schedule and the times, but please — uh, I would ask you to do also your best to respect the 15 minutes break and be here 30 seconds before the bell — rings and so on. Because otherwise we lost two minutes that of course are not subtracted from the exam, it just that I rush some proof, right? So, it's — is in your interest to be on time.

9.1 Basics of Metric Spaces

9.1.1 The Euclidean space $\mathbb{R}^n$

Spoken Protocol [00:17:15 - 00:19:02]

So, $\mathbb{R}^n$ we said before is the set of n-tuples of real numbers, right? *[Writes on the board]* And we will be concerned with $\mathbb{R}^n$, n in $\mathbb{N}$ is the dimension of the space. Can you see from up there this? Yeah? You have very good eyes because I could not see anything. Okay. Now, $\mathbb{R}^n$ by itself — well, — just vectors of — of numbers of real numbers is not — that interesting, we will need to add some extra structure. So, what is the Euclidean — Euclidean structure? You already know, I think, but we will say again.

Definition of $\mathbb{R}^n$

Definition 9.1 (Euclidean Structure of $\mathbb{R}^n$)**.** The space $\mathbb{R}^n$ is the set of all n-tuples of real numbers:

$$\mathbb{R}^n = \{x = (x_1, \ldots, x_n) \mid x_i \in \mathbb{R}\}$$

The natural number $n \in \mathbb{N}$ denotes the **dimension** of the space.

Spoken Protocol [00:19:02 - 00:21:18]

So, in $\mathbb{R}^n$ what can we do? We can measure — we can, well, sum vectors, right? *[Writes the sum formula]* We can multiply — this is like linear algebra. If λ is a real, we can multiply by a — a scalar. This is a scalar, a real number. So, $\lambda x_1, \ldots, \lambda x_n$. You know that from linear algebra very well. And then we can also measure the Euclidean structure means that we can also measure distances between points, right? So, we have the norm, the Euclidean norm. *[Writes the norm formula]*

Vector Space Structure of $\mathbb{R}^n$

The space $\mathbb{R}^n$ is equipped with the standard operations of a vector space:
Vector Addition: For $x, y \in \mathbb{R}^n$:

$$x + y = (x_1 + y_1, \ldots, x_n + y_n)$$

Scalar Multiplication: For $x \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n)$$

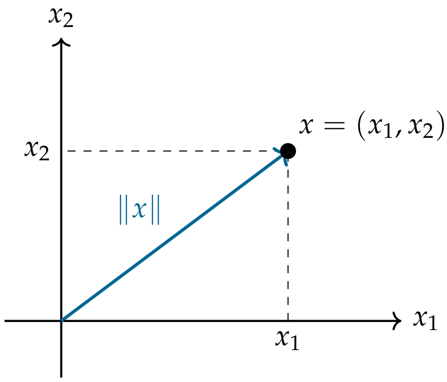
Spoken Protocol [00:21:18 - 00:22:37]

So, first the norm, the Euclidean norm of a vector x is the root, the square root of the square — the sum of the squares of the components. x_1^2 to — x_n^2. Right? *[Writes the formula on the board]* This is the length of the vector. So, in the plane, let's say... *[Draws a 2D coordinate system]* Let's say this is x_1 and this is x_2, this is the point x. The length of — of this vector x, the length, is $x_1^2 + x_2^2$ square root. Pythagoras. Right? Pythagoras theorem. And the same applies to every dimension.

The Euclidean Norm

The Euclidean norm (or length) of a vector $x \in \mathbb{R}^n$ is defined as:

$$\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2} = \sqrt{x_1^2 + \cdots + x_n^2}$$



Explanation of Steps: The Euclidean norm is a direct generalization of the Pythagorean theorem to n dimensions, representing the straight-line distance from the origin to the point x.

Spoken Protocol [00:22:37 - 00:24:42]

Okay. And we can also measure scalar products or — inner products that have, you know, maybe that they have to do with angles, right? The scalar products. So, we will use these notations: $x \cdot y$ or sometimes this also, $\langle x, y \rangle$. Both are fine. *[Writes the summation formula]* This should not be too surprising. Scalar product. And then I said the norm of a vector, that is the length of the vector, and then the distance between points. What is the distance between points? If I take two points here... *[Draws two points x and y in space]* One point x and one point y over here, the distance between the two points is the length of the vector that joins them, no? The distance between x and y is the length of the vector that joins them. This a definition. This is the Euclidean distance.

Scalar Product and Distance

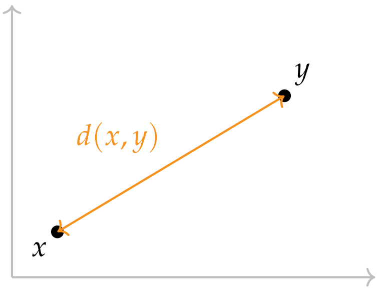
The space is further structured by the **scalar product** (or inner product):

Scalar Product

$$x \cdot y = \langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$$

The **Euclidean distance** between two points $x, y \in \mathbb{R}^n$ is defined as the norm of their difference:

$$d(x, y) = \|y - x\| = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2} \tag{9.1}$$



Spoken Protocol [00:24:42 - 00:26:37]

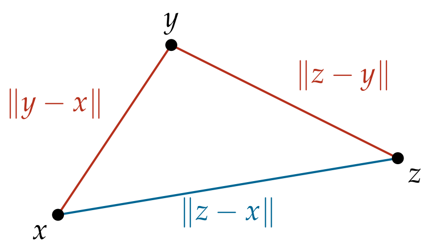
Now, the first lemma I will prove about Euclidean distance is the triangle inequality. Triangle inequality. Do you know the — some of you know the triangle inequality? Have you seen that? Who — how many people have not seen the triangle inequality? Ah, you all know the triangle inequality. Okay, then I will be quick on that. So, the triangle inequality says this: for all x, y, z in $\mathbb{R}^n$, for all — triples of vectors, if I do $z - x$, the length of this is less than the length of going from x to y plus the length of going from y to z. Right? *[Writes the inequality on the board]* Maybe you have seen the triangle inequality with the absolute value? Have — but you have also seen that one, with the modulus, with the length. Right? So, what this says is that whenever you have three points x, y, z in the space in any position... *[Draws a triangle with vertices x, y, z]* The length — so the — the distance between x and z is always longer than this — the sum of these two distances. Right? Or equal. And actually it is equal when the point is aligned with the two, right? But I'm not saying this yet.

Lemma: The Triangle Inequality

Triangle Inequality

Proposition 9.2 (Triangle Inequality in $\mathbb{R}^n$). For all $x, y, z \in \mathbb{R}^n$, the following inequality holds:

$$\|z - x\| \leq \|y - x\| + \|z - y\| \tag{9.2}$$



Explanation of Steps: Geometrically, the triangle inequality states that the shortest path between two points is a straight line. Any detour through a third point y must result in a total distance that is greater than or equal to the direct distance.

Spoken Protocol [00:26:37 - 00:29:18]

So, let's prove that. So, to prove that equivalently I will prove — equivalently, I'm saying that $\|x+y\|$ is less than or equal to $\|x\|+\|y\|$. [*Writes the equivalent form on the board and labels it with a red star*] So, why is this equivalent to this one? Well, equivalent is means that if you know how to prove this, easily you prove that, right? Why is this equivalent to this one? Well, you see the difference of the theorem, it's clear. So, what would be a proof of this? So, you need to do this substitution. You can call a the vector $y-x$, say, then b is equal to $z-y$, and then automatically if you sum $a+b$ is — is $z-x$. Right? [*Writes the substitution on the board*] This is like putting — so one — so if you want to go from here to here, you just put as he said z equal to — so you need to put here is $x+y$. Well, you need to do this substitution, this kind of substitution, and it works both ways, right? Because if you do this, you will get $a+b$ less than or equal to to that and — and the opposite. If you — use this inequality with this substitution, you will get that. Now, this is as detailed as I will be, again. A bit the difference between Analysis 1 and 2. If this equivalence poses some trouble, it shouldn't pose too much, but you can work a bit by yourself, is a nice exercise. Okay? And if you cannot, um, ask later.

Equivalence of Triangle Inequality Forms

The statement in Equation 9.2 is equivalent to the following simpler form:

Equivalent Form

$$\|x+y\| \leq \|x\|+\|y\| \qquad (*)$$

Proof of Equivalence: To see the equivalence, we perform a change of variables. Let:

$$a = y - x$$
$$b = z - y$$

Summing these vectors yields:

$$a+b = (y-x)+(z-y) = z-x$$

Substituting these into Equation 9.2, we obtain:

$$\|a+b\| \leq \|a\|+\|b\|$$

which is exactly the form of Equation $*$. Since the mapping $(x,y,z) \mapsto (a,b)$ is surjective, the two statements are logically equivalent.

Spoken Protocol [00:29:18 - 00:31:17]

Okay, so this is the — since we convinced ourselves that the two statements are equivalent and this is shorter, I will prove this one. So, now the goal is to prove that inequality. Okay? And now I will do it maybe in the next blackboard. [*Moves to the next chalkboard*] So, I start writing... let me call this inequality star. [*Writes "*" next to the formula*] So, this what I want to show is equivalent after squaring, right? I square and I said $\|x+y\|^2$ is equal — is less than or equal to $\|x\|^2+\|y\|^2+2\|x\|\|y\|$. Right? [*Writes the squared inequality*] That inequality is equivalent to this one because is an inequality between positive numbers and the square — squaring is a monotone function. This is the part that I said in words but I will not write because we are in Analysis 2. Okay? This you learned in Analysis 1. Now, once I wrote it in this form, I can develop this. What is the modulus of squared? This is equal by definition to the sum of $(x_i + y_i)^2$ between i equal 1 and n. Right? *[Writes the summation]* And this is — and this I can go — this is equal, develop the — now these are real numbers, so I can develop the squares. This is the sum... *[Audio cuts off abruptly]*

Proof Strategy for the Triangle Inequality

We aim to prove Equation $*$. Since both sides are non-negative, it is equivalent to prove the squared version:

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|$$

Expanding the left-hand side using the definition of the Euclidean norm:

$$\begin{aligned}\|x + y\|^2 &= \sum_{i=1}^{n}(x_i + y_i)^2 \\ &= \sum_{i=1}^{n}(x_i^2 + 2x_iy_i + y_i^2) \\ &= \sum_{i=1}^{n} x_i^2 + 2\sum_{i=1}^{n} x_iy_i + \sum_{i=1}^{n} y_i^2 \\ &= \|x\|^2 + 2(x \cdot y) + \|y\|^2\end{aligned}$$

[Audio cuts off abruptly]

Spoken Protocol [00:00:00 - 00:01:24]

Right? Because if you do this (i.e., the substitution $a = y - x$ and $b = z - y$), you will get $a + b$ less than or equal to to that and — and the opposite. If you, uh, use this inequality with this substitution, you will get that. Now, this is as detailed as I will be, again. A bit the difference between Analysis 1 and 2. If this equivalence poses some trouble, it shouldn't pose too much, but you can work a bit by yourself, is a nice exercise. Okay? And if you cannot, um, ask later.
Okay, so this is the — since we convinced ourselves that the two statements are equivalent and this is shorter, I will prove this one (i.e., Equation $*$). So, now the goal is to prove that inequality. Okay? And now I will do it maybe in the next blackboard. *[Moves to the center-left chalkboard]* So, I start writing... let me call this inequality star. *[Writes "*" next to the formula]* So, this what I want to show is equivalent after squaring, right? I square and I said...

Proof of the Triangle Inequality (Initial Attempt)

To prove the triangle inequality, we consider the squared norm of the sum of two vectors. *[The lecturer initially writes the formula for the difference $x - y$ on the board]*

$$\|x - y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \quad \text{(Incorrect notation)} \tag{9.3}$$

Expanding the left-hand side by the definition of the Euclidean norm:

$$\sum_{i=1}^{n}(x_i - y_i)^2 = \ldots$$

Spoken Protocol [00:01:24 - 00:02:15]

...I square and I said x minus y squared (i.e., actually $x+y$ as intended for the proof) is equal — is less than or equal to x squared plus y squared plus two times the product of the modulus. Right? That inequality is equivalent to this one because is an inequality between positive numbers and the square — squaring is a monotone function. This is the part that I said in words but I will not write because we are in Analysis 2. Okay? This you learned in Analysis 1. Now, once I wrote it in this form, I can develop this. What is the modulus of squared? This is equal by definition to the sum of x_i minus y_i squared between i equal 1 and n. Right?

Student Spotting the Sign Error

Um, shouldn't it be $x+y$ at the beginning of the proof? There [points at the board].

Spoken Protocol [continued]

Ah, yes. Actually is the same but, yeah, now that you say it, is better. So, you can put $x+y$ is — is equivalent to put $x+y$ and $x-y$ but — but to match this inequality (i.e., Equation $*$) otherwise it would be with a minus. But all the proof you can do with the plus and with the minus but — okay, then here will be a plus. Thank you. Thank you for the — nice. *[Corrects the board]*

Proof of the Triangle Inequality (Corrected)

We aim to prove Equation $*$: $\|x+y\| \leq \|x\| + \|y\|$. Since both sides are non-negative, this is equivalent to proving:

$$\|x+y\|^2 \leq (\|x\| + \|y\|)^2 = \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \tag{9.4}$$

Expanding the left-hand side using the definition of the Euclidean norm and the properties of real numbers:

$$\begin{aligned}
\|x+y\|^2 &= \sum_{i=1}^{n}(x_i + y_i)^2 \\
&= \sum_{i=1}^{n}(x_i^2 + y_i^2 + 2x_iy_i) \\
&= \underbrace{\sum_{i=1}^{n} x_i^2}_{\|x\|^2} + \underbrace{\sum_{i=1}^{n} y_i^2}_{\|y\|^2} + 2\underbrace{\sum_{i=1}^{n} x_iy_i}_{x\cdot y} \\
&= \|x\|^2 + \|y\|^2 + 2(x \cdot y)
\end{aligned}$$

Substituting this back into Equation 9.4, we see that the inequality holds if and only if:

$$\|x\|^2 + \|y\|^2 + 2(x \cdot y) \leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|$$

Canceling the common terms $||x||^2$ and $||y||^2$ and dividing by 2, we find that the triangle inequality is equivalent to the following condition:

$$x \cdot y \leq \|x\|\|y\| \tag{9.5}$$

Spoken Protocol [00:03:28 - 00:05:50]

Right? Which is equal to the modulus of x squared plus the modulus of y squared. This is the modulus of x squared when you sum over n, over i. This is the modulus of y squared and this is two times the scalar product. Right? So, I want to prove — this is a bit like a snake, but these are equalities. This is equal to this, is equal to this. So, what I'm left with — I'm left with proving that this thing is less than or equal to this thing. And you see here, this — this I can simplify, right? I can simplify because it's both sides. So, what I want to show is that two x dot y is less than or equal than the product of the modulus. Right? So, after this calculation, what I want to prove is equivalent to this. Let me simplify also the two. Is equivalent to this (i.e., Equation 9.5).
And this is an inequality that is famous. Now I will prove it in case you don't know. But this is the Cauchy-Schwarz inequality. Okay? And this is true, this inequality is true because is the Cauchy-Schwarz. *[Writes "true Cauchy-Schwarz inequality" on the board]* And now, in case just to warm up and in case you didn't remember or you never saw the proof of the Cauchy-Schwarz inequality, I will prove the Cauchy-Schwarz inequality, right? So, if I can prove this inequality, you see that the proof of the triangle inequality is done, is concluded. I'm just left with proving this inequality that I told you is called the Cauchy-Schwarz inequality. So, let me do this in a — so I will put a square that means end of the proof but up to this ingredient that I will prove next.

Lemma: Cauchy-Schwarz Inequality

Lemma 9.3 (Cauchy-Schwarz Inequality in $\mathbb{R}^n$). *For all $x, y \in \mathbb{R}^n$, the following inequality holds:*

$$x \cdot y \leq \|x\|\|y\| \tag{9.6}$$

Proof of the Cauchy-Schwarz Inequality

The first observation is that if either $x = 0$ or $y = 0$, the inequality is trivially satisfied as $0 \leq 0$. Therefore, we assume without loss of generality (WLOG) that both $x \neq 0$ and $y \neq 0$.
Recall the definition of the scalar product:

$$x \cdot y = \sum_{i=1}^{n} x_i y_i$$

We introduce a parameter $\lambda > 0$. For any such λ, we can multiply and divide each term by λ:

$$2(x \cdot y) = \sum_{i=1}^{n} 2(x_i \lambda) \left(\frac{y_i}{\lambda}\right)$$

Using the elementary inequality for real numbers $2ab \leq a^2 + b^2$ (which follows from

$(a-b)^2 \geq 0$), we apply this to each term in the sum:

$$\begin{aligned} 2(x \cdot y) &\leq \sum_{i=1}^{n} \left((x_i\lambda)^2 + \left(\frac{y_i}{\lambda}\right)^2 \right) \\ &= \lambda^2 \sum_{i=1}^{n} x_i^2 + \frac{1}{\lambda^2} \sum_{i=1}^{n} y_i^2 \\ &= \lambda^2 \|x\|^2 + \frac{1}{\lambda^2} \|y\|^2 \end{aligned}$$

This inequality holds for any $\lambda > 0$. To obtain the tightest bound, we choose λ such that the two terms on the right are equal. Setting $\lambda^2\|x\|^2 = \frac{1}{\lambda^2}\|y\|^2$ yields:

$$\lambda^2 = \frac{\|y\|}{\|x\|}$$

Substituting this optimal value of λ^2 back into the inequality:

$$\begin{aligned} 2(x \cdot y) &\leq \left(\frac{\|y\|}{\|x\|}\right) \|x\|^2 + \left(\frac{\|x\|}{\|y\|}\right) \|y\|^2 \\ &= \|y\|\|x\| + \|x\|\|y\| \\ &= 2\|x\|\|y\| \end{aligned}$$

Dividing by 2, we arrive at the Cauchy-Schwarz inequality:

$$x \cdot y \leq \|x\|\|y\|$$

Spoken Protocol [00:05:50 - 00:08:20]

So, now I say Lemma, Cauchy-Schwarz inequality. Well, I don't copy. Lemma, this holds. Right? Cauchy-Schwarz inequality. In the notes will be as a separate lemma. But now I don't make me copy again because I'm going to write again exactly the same. So, this inequality holds for all x and y in $\mathbb{R}^n$. Okay? Now, proof. In the notes I have put two proofs. There are many ways to prove this. I have put two proofs of this inequality. I will with give one, okay?

The first observation is that if either x equal zero or y equal zero, the inequality is — zero less than or equal to zero, right? Because if — if the vector is — if the vector one of the two vectors is zero, this product is zero and the modulus of one of the two will be zero so is zero less than or equal to zero. Which is trivially true. By this observation, I can assume without loss of generality that both vectors are not zero. So, assume without loss of generality (WLOG) both x and y are different from zero. This we have seen many times now you are in the second semester, this *"assume without loss of generality"* phrasing.

Spoken Protocol [00:08:20 - 00:10:30]

Okay, so both vectors are non-zero. Now what I will do, um, I will take — want to prove — I will write what is the scalar product x dot y. This is the sum between i equal 1 and n, I copy again, $x_i y_i$. Right? And now I will do a trick. Um, I will take for every — for every number λ positive, I have the right to divide and multiply by λ. I will put λ here and λ over here. Right? I did nothing because I multiplied and divided by λ. And now I — I will use an n times an inequality of real numbers. You know that — or let me — let me multiply by two here and here. And now I will use an inequality of real numbers: $2ab$ is less than or equal to $a^2 + b^2$. Which is the same as — essentially that $(a-b)^2$ is non-negative. So, this inequality of real numbers, I will apply n times to each of these two numbers. This will be the a and this will be the b in each case. And what do I get? And with the two, right? I get this is less than or equal than the sum i equal 1 to n. Now, the λ, I get $\lambda^2 x_i^2$ plus y_i^2 over λ^2. Okay? Which is the same as λ^2 mod x squared plus 1 over λ^2 mod y squared. And this holds for every λ. Right? This inequality holds for every λ positive. I didn't use anything. Now is where I use that both are — are not zero, so I can take λ equal — or λ^2 equal what? y divided by x. We take that λ and you get λ^2 two times, you see this — when you get the 1 over λ exactly simplifies what you want and you get two times the product. Okay? This is the proof of the Cauchy-Schwarz and this ends the proof of the triangle inequality. Is there any question about this proof?

Segment Transition

The lecturer has finished the proofs of the triangle inequality and the Cauchy-Schwarz inequality. He erases the center and right chalkboards to transition to the next major topic: Metric Spaces.

9.1.2 Definition of metric Space

Spoken Protocol [00:10:30 - 00:12:50]

This is the structure $\mathbb{R}^n$ with this structure of the scalar product, the length, is the main structure we will use in the proof to do analysis. But there are — now we need to understand convergence sequences in $\mathbb{R}^n$, we need to understand Cauchy sequences and all these things that you understood in $\mathbb{R}$. Everything — a bit the summary — everything that you learned how to do in $\mathbb{R}$, now we need to learn how to do in $\mathbb{R}^n$. But — to make it more efficient, we will learn all these concepts in a more general, um, — from a more general viewpoint, that is from the viewpoint of metric spaces. Right? So that later when you modify — so when you go to other — courses or in this course later on we will have different metric spaces, I don't need to redefine all the time what convergence means and so on.

Now, I'm — I need to warn you, especially for the — well, I don't know if especially for the physicists, but this — this — now will get a bit more abstract. But bear with — then we will come — this will be for a — for a couple of weeks or so. Later we will come back to $\mathbb{R}^n$, everything $\mathbb{R}^n$. But this is metric spaces topology is more axiomatic, more abstract. It's good to practice proofs because here you need to use axioms. You cannot use your intuitions all the time. You need to go back to the axioms of what is a metric space that I will tell you now and do the proofs. Sometimes working like mathematicians in more abstract viewpoint with few — fewer axioms, um, is also easier. Well, depends, right? But it's easier because if you have less axioms — now in $\mathbb{R}^n$ we have many axioms, we have the axiom of the supremum for each coordinate, you have many things. Now in the metric space, I will introduce three axioms. Just three. And then all the proofs need to use these three axioms. So, you don't need to think too much what are — it's like — like playing chess, I say sometimes, like playing chess only with the rooks. So, if you play chess only and there are only the rooks, it's not that difficult than playing with many pieces, right? So, but then you need a bit more of abstraction because it's not what you are used to. So, we are going to play chess with the rooks and we will define a metric space.

***Insight:* The *"Chess with Rooks"* Analogy**

The lecturer uses a playful analogy to explain the pedagogical value of abstraction. By stripping away the specific properties of $\mathbb{R}^n$ and focusing on the minimal axioms of a metric space, the *"rules of the game"* become simpler but require more rigorous adherence. This prevents students from relying on geometric intuition that might fail in more exotic spaces (like function spaces).

Definition: Metric Space

Definition 9.4 (Metric space)**.** A **metric space** is a pair (X, d), where X is a non-empty set and d is a mapping (called the **distance** or **metric**):

$$d : X \times X \to [0, \infty)$$

that satisfies the following three axioms for all $x, y, z \in X$:

(a) Definiteness: $d(x, y) = 0 \iff x = y$.

(b) Symmetry: $d(x, y) = d(y, x)$.

(c) Triangle Inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

Explanation of Steps: The lecturer notes that while he formally defines X as a set, he will not be overly pedantic about the case where X might be empty, as it is trivial and lacks mathematical interest for the course.

Spoken Protocol [00:12:50 - 00:15:15]

So, we will define a metric space is a pair (X, d) where X is a non-empty set. Well, let's say — this also the kind of things that I will not be that careful about. So, if the set X is empty or non-empty — I mean, this is the part that I'm sorry to say that, but who cares? Uh, if the set is — so if someone wants to say *"no, because with the empty set"*, okay, in the notes probably everything is okay and I say that the set of is non-empty. In the lecture I will not focus on the case that the set can be empty or not, unless it is important and — and hides some important subtlety. So, X is a set that is non-empty, but let's not put it. And d — this is the definition, sorry, this is the definition of metric space. And d is a map from the product X times X to the non-negative real numbers that will be called the distance. This is called the distance. And this distance satisfies three axioms.

Axiom number one is that is called is definite. That is for every pair of points x and y in X, in my space, so the distance between the two points is zero if and only if the points are the same. The distance between a point and itself is zero, of course, because what sort of distance would be non-zero from a point to itself? And if the points are different, the distance will not be zero because otherwise this distance would not be able to distinguish points, right? Axiom number two: the distance is symmetric. The distance between x and y is the same as the distance between y and x. Okay? This axiom is very also very reasonable for a distance. In applications, uh, this is not always true if you think of a distance like the distance of flying — uh, with a plane between two cities. You know that when you fly to the west is usually take less time than fly to the east? No, flying to the west takes less time than flying to the east? Due to the — to the atmospheric drift. So, it's not always true in every situation that the distance is symmetric, but in many situations it is true. And the third axiom is the triangle inequality.

Spoken Protocol [00:15:15 - 00:17:15]

These are the axioms of a metric space. First example — now I will give some examples, no? First example of a metric space: the Euclidean space $\mathbb{R}^n$ with the distance the length of a vector. Is very easy that it satisfies the — the three axioms. So, examples. $\mathbb{R}^n$ with the Euclidean distance $d_{\text{Euclidean}}$. Why do we specify both the set and the distance? Because sometimes the same set has two or more interesting distances, right? For instance, in $\mathbb{R}^2$ I can put the Euclidean distance or I can put a different distance that you can check also satisfies that is called the — in $\mathbb{R}^2$ there is this distance, uh, the New York distance (i.e., the Manhattan distance). That is mod x_1 minus y_1 plus mod x_2 minus y_2. Right? This is called the Manhattan distance. Why? Because this is the sort of distance when you live in one of these modern American cities or — where the streets go north to south and east to west and you want to go from point x to y, right? Then you cannot go diagonal because you hit the buildings and then you need to go like that. And then the distance between two points is — is this thing. Okay? And you can check very easily that this still satisfies all the axioms of the distance. Right?

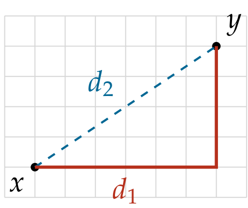
Examples of Metrics on $\mathbb{R}^n$

(a) **Euclidean Metric:** The standard distance in $\mathbb{R}^n$:

$$d_2(x,y) = \|x-y\| = \sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}$$

(b) **Manhattan Metric (Taxicab Metric):** Common in grid-based layouts:

$$d_1(x,y) = |x_1-y_1| + |x_2-y_2|$$



Spoken Protocol [00:17:15 - 00:19:15]

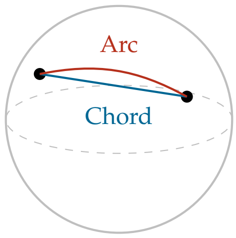
So, when you have a bit more abstract example, when you have any metric space and you have a subset of it, you can just restrict this distance to the — to Y. So, d restricted to Y cross Y with Y, this is another metric space. Right? If you take a subset of your given metric space and restrict the distance, is — will be always a metric space. So, example, more — this is a bit always true, a more concrete example would be if this is $\mathbb{R}^3$ with the Euclidean distance, right? And I take a subset that is the two-dimensional sphere. Let me draw it like this. S^2 is the set of points x in $\mathbb{R}^3$ such that the length of x is one. Right? This S^2, the sphere. Then I can see the sphere as a metric space with the Euclidean distance. So, the distance between these two points of the sphere, this one and this one say, is just you take the straight segment and measure the length. And this satisfies the axioms of the distance. Is a restriction of the Euclidean distance. You can say *"okay, but this distance maybe is not the best"*. Because as I was saying before, if you travel by plane, this is not the distance between points. The distance between points — a first approximation of the distance between points when you travel by plane is the geodesic distance. That is the length of the arc, the shortest arc on the surface of the earth. This is another distance. Right? Same set can have multiple meaningful distances depending on the context.

Subspaces and the Sphere

Any subset $Y \subset X$ of a metric space (X, d) inherits a metric structure by restricting the distance function to $Y \times Y$.
Example: The Unit Sphere $S^2 \subset \mathbb{R}^3$:

$$S^2 = \{x \in \mathbb{R}^3 \mid \|x\| = 1\}$$

The sphere can be equipped with two distinct metrics:
Inherited Euclidean Metric: The straight-line distance through the interior of the sphere.
Geodesic Metric: The distance measured along the surface of the sphere (great-circle distance).



Spoken Protocol [00:19:15 - 00:21:15]

And now maybe before — let me do a couple more of examples. I will put it here. Maybe I need to erase something. So, one thing that can be useful when you try to do a proof in this setup that is a bit abstract of metric spaces, remember you have very few axioms you can use, but then you have less intuition. So, how you — how can you overcome your lack of intuition? What you do is you — you think of a specific example. Don't think of a general metric space. A general metric space means a billion of things at the same time, right? Is an abstract model of many things at the same time. Pick a concrete example. Pick the Euclidean space. Do your proof in the Euclidean space using your intuitions that you have from the space. And then try to see which axioms are you using. If you can use — maybe you use the wrong proof and you are not using just those axioms. But then try to modify your proof using your intuition until you can do with those axioms. Okay? This can be useful. So, whenever things are abstract, don't think always in the abstract, just go to concrete examples. And this applies always because I have seen many times in more advanced courses doing examinations that you ask — so people know the abstract theory very well and then you ask in a concrete case. *"So, give me the — no, no, don't tell me the Ricci curvature on a — no, what is in the Euclidean space?"* And then people sometimes don't know how to answer. And this is a problem. So, the abstract is good but you need to know very well and think — spend time thinking about concrete examples.

So, let me give you so that you can see also that things start to be interesting very soon. Let me give you some more interesting examples of metric space, right? Now we saw spaces of points. We can also consider spaces of functions. Now, you learned what is a continuous function. Let's say a, b are two real numbers, $a < b$. And you can take X the space of continuous functions from the closed interval $[a, b]$ to $\mathbb{R}$. f continuous. This is a space of functions. So, one point in this space would look something like this... *[Draws a curve on a coordinate system]* This is one point in this space. How can we define a distance between functions? Well, in many ways. Something that is used often is the supremum distance or the maximum distance. Is the distance between f and g is the maximum in $[a, b]$ of the absolute value of the difference. So, you take two curves here, different, and then you look at the maximal oscillation — so the maximum difference between the two functions. This is a distance between functions. If this is zero, the two functions are equal? Yes. Only if? Yes, is a bit more you need to reason but they are continuous so you can do that. Exercise. Does it satisfy the triangle inequality? Prove it. The symmetry is very easy that you replace f and g is the same because here you have the absolute value. But the triangle inequality you can prove it using the triangle inequality for the absolute value and playing with the maximum.

Another distance between functions... *[Writes the integral formula]* You measure the distance, you square, you integrate and you take the square root. This is another distance. Very important, the L^2 distance. We will not use this in a course, but you will see that you will use this later on many, many times. So, we will continue next time from — from here.

Metrics on Function Spaces

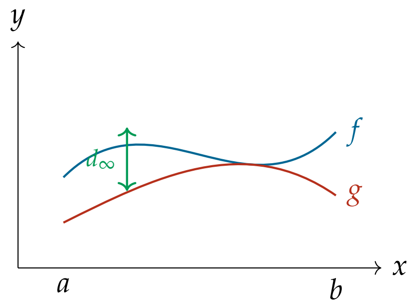
Let $X = C([a, b], \mathbb{R})$ be the set of all continuous functions $f : [a, b] \to \mathbb{R}$. We can define several metrics on this set:

(a) Supremum Metric (L^∞ Metric):

$$d_\infty(f, g) = \max_{x \in [a,b]} |f(x) - g(x)|$$

(b) L^2 Metric:

$$d_2(f, g) = \left(\int_a^b (f(x) - g(x))^2 \, dx \right)^{1/2}$$



Explanation of Steps: The L^∞ metric measures the *"worst-case"* difference between two functions, while the L^2 metric provides an *"average"* measure of their difference over the entire interval. Both satisfy the metric axioms, demonstrating that the concept of distance extends far beyond simple points in space.

CHAPTER 9

Wednesday: February 18th 2026

Metric Spaces: Sequences and Completeness

Context

This lecture continues the exploration of Metric Spaces, transitioning from the foundational axioms of distance to the rigorous study of sequences, convergence, and accumulation points. The lecturer emphasizes the continuity of concepts from Analysis 1, now generalized to arbitrary sets equipped with a metric.

9.1 Basics of Metric Spaces

9.1.3 Sequences, limits, and completeness

Spoken Protocol [00:00:00 - 00:00:50]

So, uh, hopefully by the second week it will all take us two seconds to go from chatting, uh, to full silence. So, uh, we will continue with metric spaces, provided my voice allows. Uh, remember last time we introduced this pair *[points at the board]*: (X, d). X is a set and d was a map from $X \times X$ to positive numbers, or non-negative numbers. And now for today, you can forget about the rest of math you learned, because we are going to use only this and the three axioms. So, you just need to remember three pieces of information: the three axioms.

Foundational Metric Space

We operate within a metric space defined by the pair:

$$(X, d)$$

where X is a set and $d : X \times X \to [0, \infty)$ is a distance function satisfying the three fundamental axioms (identity of indiscernibles, symmetry, and the triangle inequality).

Definition of a Sequence

Spoken Protocol [00:00:50 - 00:01:50]

So, uh, now we will learn again concepts you, uh, you saw in Analysis 1, and the first concept is that of sequence. *[Writes on the board]* Definition: sequence. You saw sequences of real numbers, uh, so if X is a set, we define — we call sequence in X a map that we put like this: x from the natural numbers to X. This is small x, this is big X, okay?

Definition: Sequence

Definition 9.15 (Sequences in a set)**.** Let X be a set. A **sequence** in X is a mapping from the set of natural numbers $\mathbb{N}$ to the set X:

$$x : \mathbb{N} \to X$$

Spoken Protocol [00:01:50 - 00:03:00]

And as in Analysis 1, instead of denoting this map $x(n)$ as we do with functions, we will use these notations: x_n, right? Sub-index. And when we will — when we want to denote the full map, the — this is just one element, the n, the n-th element of the sequence. This is the n-th element that belongs to X, belongs to the set X, capital X. And when we want to denote the full sequence, we will say: let (x_n) like this be a sequence, and then we can put this, we can put this *[writes various notations on the board]*. So, all of these are equivalent notations. Or we can put, I don't know, things like that. So, this is our notation for sequence, right?

Sequence Notation

While a sequence is formally a function, we typically employ subscript notation to denote its elements.
Individual Element: x_n represents the n-th element of the sequence, where $x_n \in X$.
Full Sequence: To denote the entire sequence, the following notations are considered equivalent:

- $(x_n)_{n\geq 0}$
- $(x_n)_{n=0}^{\infty}$
- $(x_n)_{n\in\mathbb{N}}$

Convergence and Limits

Spoken Protocol [00:03:00 - 00:04:11]

But now we are in a — in a metric space, so we can define also convergent sequence and limit. Definition number two will be: convergent sequence and limit. *[Writes on the board]* So, we say (x_n) — so this is for a metric space, so (X, d) is a metric space, d is the distance, and $(x_n)_{n\geq 0}$ has limit x in X if... well, we can put it in many ways.

Definition: Convergence and Limit

Definition 9.18 (Convergent sequence)**.** Let (X, d) be a metric space. A sequence $(x_n)_{n\geq 0}$ in X is said to **converge** to a limit $x \in X$ if the distance between the sequence elements and the limit point approaches zero:

$$d(x_n, x) \to 0 \quad \text{as real numbers.}$$

Spoken Protocol [00:04:11 - 00:05:15]

But essentially what we will say is that if we — if we have a sequence of points in our space, it converges to our given limit point if the distance converges to zero. We have the distance, we need to phrase everything in terms of the distance. So, if the distance between x_n and the limit point converges to zero as real numbers, right? These are — these are points in your abstract space, but the distance between the two points is a number. So, I can say if this distance goes to zero, uh, this means that this point is the limit of that sequence.

Insight: The Bridge to Real Numbers

The lecturer highlights a crucial conceptual bridge: while x_n and x are elements of an abstract set X, the distance $d(x_n, x)$ is always a real number. This allows us to use the familiar notion of limits from $\mathbb{R}$ to define convergence in any metric space.

Spoken Protocol [00:05:15 - 00:06:23]

So, equivalently, if you want something very similar, equivalently: for all ϵ positive, exists capital N that depends on ϵ, such that the distance between x_n and x is less than ϵ whenever n is bigger than capital N, right? This is the same, more analogous to what you have seen in Analysis 1 for sequences. Okay. But now this is super general because we saw last time that metric spaces could be a space of functions, could be anything. Now you know how to do sequence — convergent sequences in any, uh, in very general setup.

Formal ϵ-N Definition of Convergence

The convergence $x_n \to x$ is formally defined as:

$$\forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \text{ s.t. } \forall n \geq N, d(x_n, x) < \epsilon$$

Explanation of Steps: This definition mirrors the classical definition of convergence in $\mathbb{R}$, replacing the absolute difference $|x_n - x|$ with the general metric $d(x_n, x)$. It states that for any arbitrarily small distance ϵ, there exists a point in the sequence beyond which all subsequent elements are *"trapped"* within an ϵ-neighborhood of the limit x.

Spoken Protocol [00:06:23 - 00:07:38]

Okay, so, uh, now we know — well, let me once and for all give you the notations we will use. So, when the — when the ambient, when the — when the space you are using is clear, most of the time in this course will be $\mathbb{R}^n$ with the Euclidean distance, or maybe the complex numbers that is $\mathbb{R}^2$, right? Um, then we will write to say that, uh, x_n has limit x, we will use these notations: that limit in n, or when n goes to infinity, of x_n is equal to x. This we can write when — when everything is clear from the context, right? When the X and the distance. Or we can also put even shortly: x_n converges to x. This I like a lot because it is very compact. Right? As in here, I used it in here. Okay.

Limit Notation

When the metric space (X, d) is understood, we use the following standard notations for the limit:

- $\lim_{n\to\infty} x_n = x$
- $x_n \to x$

Uniqueness of Limits

Spoken Protocol [00:07:38 - 00:09:12]

Now, we define — given a convergent sequence, we define — or we said what is its limit. Now, first question: is the limit unique? Let's start for the very, very basic, uh, fundamental questions. So, first lemma. *[Writes on the board]* Oh, nice. What is this? Um, is — so, Lemma: the limit of a sequence is unique. Let's say: if — so in a metric space (X, d), right? And now you take a sequence $(x_n)_{n\geq 0}$, and you have that x_n converges to x and you know also that x_n converges to y, two points, given points of X. x, y in X. So, this is a sequence, and this is an assumption. Assume that x_n converges to x and x_n converges to y. Then $x = y$. Right?

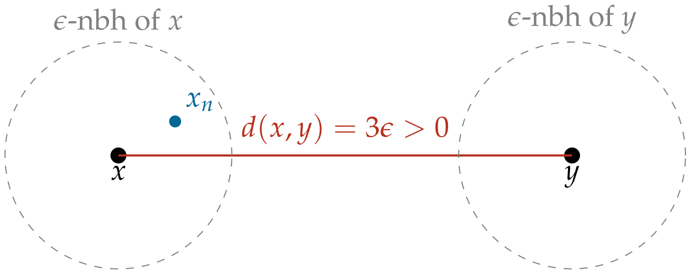
Lemma: Uniqueness of the Limit

Lemma 9.21 (Uniqueness of the limit)**.** *In any metric space (X, d), if a sequence (x_n) converges to x and also converges to y, then x must be equal to y.*

Spoken Protocol [00:09:12 - 00:11:03]

So, how do we prove this? *[Draws a sketch on the board]* So, these are two potential limit points, x and y. And remember that the first axiom of the distance says that the distance distinguishes the points. So, the distance between two points is positive. So, let me call the distance between x and y, I will call it $2 \times \epsilon$ (i.e., $d(x, y) = 2\epsilon$) that is a positive number. All right? Or to make — to make it more dramatic, I can put $3 \times \epsilon$. And also more symmetric, because the 3 and the ϵ are symmetric. Okay. Um, then this is 3ϵ. Now, the definition of convergence tells us that after a certain point — so for very, very large n — all the elements of the sequence will fall close to the limit. But at the same time, all the elements of the sequence will fall close to that limit. And this should contradict another axiom that is the triangle inequality. Okay? Here we see the intuition. So, all the elements cannot be at the same time here and here, right? So, this is what we are going to do. And now we write — we write this in symbols using the definition, right?

Intuition for Uniqueness Proof



Explanation of Steps: The proof relies on a contradiction. If $x \neq y$, their distance $d(x,y)$ is some positive value, say 3ϵ. For sufficiently large n, x_n must be within ϵ of x AND within ϵ of y. However, the triangle inequality dictates that the total distance $d(x,y)$ cannot exceed the sum of the distances $d(x,x_n) + d(x_n,y)$, which would imply $3\epsilon < 2\epsilon$—a clear contradiction.

Spoken Protocol [00:11:03 - 00:12:38]

So, by definition of the limit, exists — so given that ϵ over there, exists N — let's say N_x — such that the distance between x_n and x is less than ϵ for all n bigger than N_x, right? And the same with y: exists some N_y such that the distance between x_n and y is less than ϵ for all n bigger than or equal to N_y. And then if I take capital N to be the maximum of these two numbers, what happens? That for all n bigger than or equal to N, it is bigger than both, so I will have the distance between x and y — this was what? This by definition was $3 \times \epsilon$. Right? Now I can use the triangle inequality with these three points: x, y and x_n. So, this is less than or equal to the distance between x and x_n plus the distance between x_n and y. I'm using the triangle axiom. Now I want to have this, right? So, I need to switch these two arguments. So, I'm using the second axiom here: the symmetry. So, axiom number three, triangle inequality; axiom number two, I can switch the points in the distance. And then I get that this is less than $\epsilon + \epsilon$. So, 3ϵ is less than 2ϵ, contradiction. So, the contradiction comes from the fact that I assumed that the points are different and the distance is positive. If the points are the same and the distance is zero, there is no contradiction, of course. So, this finishes the proof of the uniqueness of the limit.

Formal Proof: Uniqueness of Limits

Assume $x_n \to x$ and $x_n \to y$ with $x \neq y$. By the first metric axiom, $d(x,y) > 0$. Let $d(x,y) = 3\epsilon$ for some $\epsilon > 0$. By the definition of convergence:

- $\exists N_x \in \mathbb{N}$ such that $\forall n \geq N_x, d(x_n, x) < \epsilon$.
- $\exists N_y \in \mathbb{N}$ such that $\forall n \geq N_y, d(x_n, y) < \epsilon$.

Let $N = \max(N_x, N_y)$. Then for all $n \geq N$, both conditions hold. Applying the triangle inequality (Axiom 3) and symmetry (Axiom 2):

$$\begin{aligned} 3\epsilon = d(x,y) &\leq d(x,x_n) + d(x_n,y) \\ &= \underbrace{d(x_n,x)}_{<\epsilon} + \underbrace{d(x_n,y)}_{<\epsilon} \\ &< \epsilon + \epsilon = 2\epsilon \end{aligned}$$

This yields $3\epsilon < 2\epsilon$, which is a contradiction since $\epsilon > 0$. Thus, our assumption $x \neq y$ must be false, and $x = y$. Q.E.D.

Spoken Protocol [00:12:38 - 00:14:14]

Um, can you — can you tell me now — how is the — how is this pace? Is it very slow? Is it very quick? Is it okay? Yeah, good. So, it's important I tell you because I have the experience from two years ago: there are some people that whatever I do, they always find it super slow, right? And these are the people that sometimes are more active looking at me and giving feedback. So, it's very important that if people start to be confused, before it's too late, I know. And in this direction, today there will be the election of the — before the break, I think it's today — there will be the election of the class speakers. And this is very useful, and I ask you to volunteer because then I have a more fluid communication. Maybe some people don't come to complain to me that I'm going too fast or too slow, but then you can go to your speakers and they will give feedback. And like this we will adjust as we go.

Student Interaction

I do not quite understand where the contradiction is. So, why this proof actually works. I do not get the last line you wrote there.

Spoken Protocol [continued]

This last line? [*points at the board*] Yes. Okay. This is the definition of 3ϵ. Now, the triangle inequality says that when you have three points, the distance between two of them is shorter than going from one to the third and to the third to the second. Right? This is the triangle. This is less than or equal to. Now, if we switch these two and we use that and that, the switching is by the second axiom, I can switch the two points. And I get less than $\epsilon + \epsilon$. So, I am getting that 3ϵ is less than 2ϵ, which is a contradiction, right? Okay. So, the only place where I could — where the contradiction came from is because I assumed that the points are different and the ϵ is positive. If the ϵ is zero, you cannot do that. Okay?

Subsequences and Accumulation Points

Definition of a Subsequence

Spoken Protocol [00:15:32 - 00:17:23]

So, next thing that — that you learned in Analysis 1 is definition: subsequence. It will be the same. Right? If (x_n) is a sequence in a set X, we define — so is there some problem or some...? You have some, yeah? No? No, but it's important — is there some confusion with this proof? I can spend more time, we have time. Yeah, good. [*The lecturer retrieves the catch-box microphone*] So, now is the tricky part, right? Almost. [*Tosses microphone to student*]

Student Interaction

Um, so can we use the terminology of limit if x isn't in the set X? So, for example, if we have like $\mathbb{R}$ but without zero and like x_n is defined as $1/n$, it converges to zero but zero isn't in X.

Spoken Protocol [continued]

No, in this context, no. Okay. So, here you — you can use because your ambient — so you are artificially removing part of your set, but you can measure the distance between zero and the positive numbers because you are in $\mathbb{R}$, and $\mathbb{R}$ has its distance. But by the definition, you see, when you look at the definition, all the things that appear in the definition need to have a meaning. And when you have the distance of x_n and x goes to zero, so that this number is a real number is well-defined, you need the both points to be in your space. Otherwise, I don't know what is the distance between something that lives in one space and something that lives in another galaxy, right? Okay.

Definition: Subsequence

Definition 9.22 (Subsequence)**.** Let $(x_n)_{n\geq 0}$ be a sequence in a set X. A **subsequence** is any sequence of the form:

$$(x_{f(n)})_{n\geq 0}$$

where $f : \mathbb{N} \to \mathbb{N}$ is a strictly increasing function.

Explanation of Steps: A strictly increasing map f ensures that we are picking elements from the original sequence in their original relative order, but potentially skipping some. Formally, f satisfies:

$$f(n) < f(m) \iff n < m$$

In standard notation, we often write n_k instead of $f(k)$, denoting the subsequence as $(x_{n_k})_{k\geq 0}$.

Accumulation Points

Spoken Protocol [00:17:23 - 00:18:44]

So, something you already learned in $\mathbb{R}$ also: accumulation points. You know what is an accumulation point. Let's put in here. Definition: accumulation points. And we will give two things, two definitions. So, if Y is a subset of X — I will not write every time, but now unless — so every time (X, d) is a metric space. And Y is a subset of X, subset. We say that y in X is an accumulation point of Y if exists a sequence (y_n) contained in Y such that y_n converges to y. Bullet point number one. And bullet point number two is for sequence. So, if (x_n) is a sequence of X, we say the x is an accumulation point of this sequence if exists a subsequence that converges to x.

Definition: Accumulation Points

Definition 9.23 (Accumulation points)**.** Let (X, d) be a metric space.

For a Set: A point $y \in X$ is an **accumulation point** of a subset $Y \subseteq X$ if there exists a sequence $(y_n)_{n\geq 0}$ with $y_n \in Y$ for all n, such that:

$$\lim_{n\to\infty} y_n = y$$

For a Sequence: A point $x \in X$ is an **accumulation point** of a sequence $(x_n)_{n\geq 0}$ if there exists a subsequence $(x_{n_k})_{k\geq 0}$ such that:

$$\lim_{k\to\infty} x_{n_k} = x$$

Spoken Protocol [00:00:36 - 00:01:43]

So, uh, you will notice that I'm going — I'm going very slowly, uh, speaking slowly, writing slowly, but I'm writing — I'm not writing very much on the blackboard. The reason — I'm giving you a lot of time to see if what I write makes sense, but I'm not writing a lot because when I start later to increase the speed and write like this [*gestures quickly*], you will need to follow, right? So, uh, if — if something is not clear because I need to write more — I say everything speaking, right? But I don't write everything on the blackboard. If someone is confused, raise your hand, stop me, we have time. Okay? But this will be very useful, for instance, I use abbreviations or things like that. So, later when I want to go a slightly quicker, this will be very useful that — that we are all used to that. Yeah.

Classroom Interaction

The lecturer retrieves the catch-box microphone to prepare for a student question.

Student Interaction

Um, is there a reason why we often use for definitions something we kind of learned in Analysis 1 as corollaries or lemmas? For example, for the accumulation point, we usually saw the definition with this ϵ-δ thing and that whenever you have this δ-neighborhood around your point and find the intersection with Y, the set is not empty. Is there a reason why we avoid ϵ-δ?

Spoken Protocol [00:02:12 - 00:03:12]

No, there is no reason. With the limit, no. Um, the ϵ-δ was more usually for continuity, right? In the continuity we will see the sequence, the ϵ-δ, we will see that everything is equivalent. Here you cannot do that much ϵ-δ. You can use ϵs or you can use limits, and in the notion of limit the ϵ is implicit. Because when I gave the definition of limit there was an ϵ — I gave with the ϵ and without the ϵ. But no, it should be exactly equivalent as in Analysis 1. When there is something like that that you have more than one possibility, as when we define the continuity of maps, of functions, I will give all the equivalent definitions systematically. But here, there are no ϵ-δs that I can see yet.

Transition

The lecturer erases the left and center chalkboards to begin a new derivation.

Characterizing Convergence via Accumulation Points

Spoken Protocol [00:03:13 - 00:05:03]

Uh, let me prove something. Easy lemma. So, a sequence — this is to emphasize also that a sequence in general is not convergent. So, it will have several accumulation points. Many. So, when a sequence is convergent, it is exactly when the sequence has only one accumulation point. And this is something you know for — for real numbers, and now we see it is the same in a metric space.

Lemma: Convergence and Accumulation Points (Initial State)

Board Action

The lecturer writes the first version of the lemma on the board.

Lemma. (Incorrect Statement)
A sequence $(x_n)_{n\geq 0} \subset X$ in a metric space (X, d) converges to x if and only if x is its sole accumulation point.

$$x_n \to x \iff x \text{ is its only acc. pt.}$$

Spoken Protocol [00:05:04 - 00:06:50]

So, let's prove that. Proof. [*Writes on the board*] So, what we need to show is that, um, x_n converges to x is equivalent — I will put it in this phrasing — x_n converges to x is equivalent to: for every accumulation point — it is a limit of a subsequence, right? For every convergent subsequence (x_{n_k}) converging to y for some y, is $y = x$? Right? I'm rephrasing this and I'm rephrasing this. I'm saying these two things are equivalent. Now, there should be one implication that is easy, right? Because I can take x_{n_k} could be x_n, right? I could take the — the full sequence is always a partial subsequence of the — it's the case that f is the identity, $f(n) = n$. Taking this, I find that x_n converges to x. So, this proves this implication [*points at the board*]. Just taking a particular case of a subsequence.

Proof of Lemma 9.1.3 (Partial)

We aim to show the equivalence:

$$x_n \to x \iff (\forall (x_{n_k}) \text{ s.t. } x_{n_k} \to y \implies y = x)$$

The implication ($\Longleftarrow$) is considered trivial by the lecturer: If every convergent subsequence must converge to x, then since the sequence (x_n) is a subsequence of itself (where $f = \text{id}$), it must also converge to x. Q.E.D.

Student Interaction

In the first implication we show to the left... how do we know x_n is convergent? Because we assume that any convergent subsequence converges to x, but we do not know that x_n is convergent a priori.

Spoken Protocol [00:08:16 - 00:09:30]

So, you say... ah, okay. So, every convergent subsequence converges to x. Uh, maybe I phrased the lemma... I wanted to... so, now you are asking me a — so, let me think again. A sequence x_n converges to x... yes, I said that x is the only accumulation point. Yeah. And you are telling me... so, if I take a subsequence that converges to y, I know that y is x. Good. But you are telling me why the full sequence should be a convergent subsequence? Good. Good question. Well spotted. Um, yes, I had in mind a slightly weaker result and then I stated... this is also true, but then this implication is a bit harder. Okay. Very good, very good. We can discuss.

Insight: **The "False" Lemma**

The lecturer intentionally (or accidentally) states a lemma that is subtly incorrect in general metric spaces without further assumptions like compactness. This creates a *"circus"* moment where a student spots a logical gap, forcing a more rigorous re-evaluation of the relationship between accumulation points and convergence.

Spoken Protocol [00:09:31 - 00:11:23]

Okay, so this is a — yeah. Can you throw there? Yeah, let's — let's have all the questions and then we will try to solve all of them. Okay. It's a very good discussion. Yeah. [*Tosses microphone to another student*]

Student Interaction

So, what would happen if you take a sequence that for even n is just n, and for odd n is $1/n$? Because then you have an accumulation point at zero, but the entire sequence has no limit. What happens then?

Spoken Protocol [continued]

[*The lecturer pauses to think, looking at the board*] Okay. Let me — let me ask this question out loud, okay? Because... so, you say, okay, we have $\mathbb{R}$ [*writes on the board*]. And you take this sequence that for n equal — so for, let's say, $x_{2n} = n$ and then $x_{2n+1} = 1/n$, right? This is the question. And then you say this sequence satisfies that every accumulation point — you say is only zero, right? Because this is unbounded, this goes to infinity. And then every subsequence, it seems to satisfy this. Yeah. Good question.

Counter-example to Lemma 9.1.3

The student provides a counter-example in $\mathbb{R}$ to show that having a unique accumulation point does not imply convergence: Consider the sequence $(x_n)_{n\geq 0}$ defined by:

$$x_n = \begin{cases} n & \text{if } n \text{ is even} \\ \frac{1}{n} & \text{if } n \text{ is odd} \end{cases}$$

Explanation of Steps: For this sequence, the only accumulation point is 0 (from the odd terms). The even terms $x_{2n} = 2n$ diverge to infinity and thus do not contribute any accumulation points in $\mathbb{R}$. However, the full sequence (x_n) clearly does not converge to 0 because the even terms grow without bound. Thus, the condition *"sole accumulation point"* is insufficient for convergence in non-compact spaces.

Spoken Protocol [00:11:24 - 00:13:40]

Good, let's put all — any other question? We will try to solve all of them. Okay. It's a very good discussion. Yeah. I mean, in the — in the script, something is slightly different. We assume that every subsequence converges to x. We don't say that x is the only accumulation point. Which would be correct, but this is... yeah, this is the — the stronger statement I was talking about. So, what script are you looking at? At the one from two years ago? Because it says that we assume that every subsequence converges to x. Yeah, this is much easier. And actually is true. Okay. So, every subsequence converges to x.

Board Action

The lecturer writes *"FALSE"* over the first lemma and writes two new, corrected versions.

FALSE

Correcting the Lemma

Lemma (Lemma 1 - Corrected)**.** A sequence $(x_n)_{n\geq 0} \subset X$ converges to x if and only if every subsequence $(x_{n_k})_{k\geq 0}$ converges to x.

$$x_n \to x \iff \forall (x_{n_k})_{k\geq 0}, x_{n_k} \to x$$

Explanation of Steps: This version is trivially true because the sequence itself is a subsequence. If every subsequence converges to x, then the sequence must converge to x. Conversely, if the sequence converges to x, any selection of its elements (a subsequence) must also converge to x.

Spoken Protocol [00:13:41 - 00:15:50]

But what I wanted to put — Lemma 3, that is a bit more interesting, is that if x_n... so, the same assumptions. Same assumptions as above. Then x_n converges to x if and only if every subsequence (x_{n_k}) has a sub-subsequence — so a subsequence of the subsequence — such that $x_{n_{k_l}}$ converges to x. So, whenever I take a — I can choose a subsequence, maybe this subsequence does not converge. But then, we will see next day the definition of compact spaces — and you know in $\mathbb{R}$. So, in compact spaces, like here in your example, you have realized that your example, you could do this example because the space is unbounded. The $\mathbb{R}$ is unbounded. Or is not compact, right? If you restrict to a compact interval of $\mathbb{R}$, what I wrote here was true. So, we see it's not completely false, it just we need some extra assumptions. In a compact space that is complete — but you will learn this today — in a complete space that is compact, this will be true. But we need some extra assumptions. Instead, this thing — is true always. Every subsequence (x_{n_k}) has a sub-subsequence that converges to x, then the limit is x. Right? And since today I need to do other things, and this is a good — it's a bit difficult but it's a good exercise, we will do that. I will put this statement as exercise. You try to solve it, and then the next day or in a couple of days when many people have tried, I will solve it for you here. And we will discuss again. But like this you can practice a bit. It's a good exercise. It's not trivial, that one. This one is a bit easy, is the one I gave. Is easier.

Lemma 3: Sub-subsequence Characterization

Lemma (Sub-subsequence Characterization of Convergence)**.** Under the same assumptions as above, a sequence $(x_n)_{n\geq 0}$ converges to x if and only if every subsequence $(x_{n_k})_{k\geq 0}$ contains a further subsequence (a *"sub-subsequence"*) $(x_{n_{k_l}})_{l\geq 0}$ that converges to x:

$$x_n \to x \iff \forall (x_{n_k})_{k\geq 0}, \exists (x_{n_{k_l}})_{l\geq 0} \text{ s.t. } x_{n_{k_l}} \to x$$

Exercise

The lecturer marks Lemma 9.1.3 as an *"exercise"* for the students to solve before the next lecture.

Transition

The lecturer erases the right chalkboard to transition to the next major topic: Completeness.

Complete Metric Spaces

Spoken Protocol [00:15:51 - 00:17:20]

Actually, this false lemma is very appropriate because it will allow me to — so it will be a very good example to introduce two concepts. The next concept I'm going to introduce is the concept of complete space. Right? Complete space. We know that $\mathbb{R}^n$ is complete — we know that $\mathbb{R}$ is complete, sorry. Um, this we proved in Analysis 1. And now you are going to see the definition of complete space like in this abstract level, and we will prove that $\mathbb{R}^n$ is complete. Okay.

Cauchy Sequences

Spoken Protocol [00:17:21 - 00:18:48]

So, first we need to define Cauchy sequence. Cauchy sequence. *[Writes on the board]* So, it's the same definition as in $\mathbb{R}$, right? So, $(x_n)_{n\geq 0}$ is Cauchy — a sequence is Cauchy if for all ϵ positive, exists capital N such that the distance between x_n and x_m is less than ϵ whenever n and m are both bigger than N. Right? If you wait enough — so if both indices are sufficiently large, then the two — the two points, any two points of the sequence become close. Right? This is the same as in $\mathbb{R}$. And now when you know Cauchy sequence, we can define a complete — so complete space. *[Writes on the board]* We say that (X, d) is complete if every sequence — every Cauchy sequence converges to some limit in the space. Right?

Definition: Complete Metric Space

Definition 9.27 (Cauchy sequence)**.** A sequence $(x_n)_{n\geq 0}$ in a metric space (X, d) is **Cauchy** if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } d(x_n, x_m) < \epsilon \quad \text{whenever } n, m \geq N$$

Definition 9.29 (Complete metric space)**.** A metric space (X, d) is **complete** if every Cauchy sequence in X converges to a limit $x \in X$.

Spoken Protocol [00:01:25 - 00:02:26]

This fact that it converges to some limit in the space I can put in the parenthesis, because in my definition of convergence we say that the sequence converges if there is a point of the space towards which it is converging. So, whenever I say that a sequence converges, it's to some point in the space. We cannot remove points from the space. So, if we start to remove points, as your peer was asking before, what happens if I take the positive numbers? Okay, then I'm taking a complete space that is $\mathbb{R}$ and I'm making it incomplete by removing points, right?

Spoken Protocol [00:02:26 - 00:03:12]

So, here we get that if the Cauchy sequence doesn't converge, then the metric space is not complete. Exactly. But my question is: what does it mean for a Cauchy sequence to not converge? I can give you some example. Let's give some examples of non-complete spaces that you know very well. Okay?

> ***Student Interaction***
>
> *So, here we get that if the Cauchy sequence doesn't converge, then the metric space is not complete?*

Spoken Protocol [continued]

Exactly.

Transition

The lecturer erases the left and center chalkboards to prepare for examples of completeness.

Examples of (In)complete Spaces

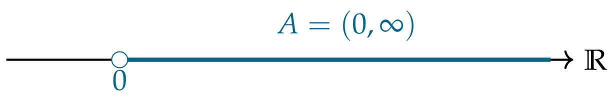
Spoken Protocol [00:04:05 - 00:05:24]

So, example of complete space we have $\mathbb{R}$, right? We have $\mathbb{R}$ with the distance — the Euclidean distance. *[Writes on the board]* What is the Euclidean distance in $\mathbb{R}$? Is the distance between two numbers is just the absolute value of the difference, right? And now we take a subset of $\mathbb{R}$. A subset $\mathbb{R}$. Then we can restrict the distance to A, the absolute value of the difference. Now, so let's say that Case number one: A equals, say, $(0, \infty)$. Is this complete?

Example: The Open Interval $(0, \infty)$

Consider the set $A = (0, \infty) \subset \mathbb{R}$ equipped with the standard Euclidean metric:

$$d(x, y) = |x - y|$$



Spoken Protocol [00:05:24 - 00:06:15]

What is the problem here? This is not complete because you are missing the zero. So, you — so this is open, right? And you can have a Cauchy sequence, the sequence $x_n = 1/n$. *[Writes on the board]* This sequence is Cauchy, and the limit would be zero, but you are removing the limit from your space. So, this space is not complete. Right?

Incompleteness of $(0, \infty)$

The space (A, d) is **not complete**.
Cauchy Sequence: Consider $x_n = \frac{1}{n}$ for $n \geq 1$.
Limit in $\mathbb{R}$: In the ambient space $\mathbb{R}$, $\lim_{n\to\infty} x_n = 0$.
Failure of Completeness: Since $0 \notin A$, the sequence does not converge to any point within the metric space (A, d).

Spoken Protocol [00:06:15 - 00:07:34]

Instead, if you take a closed set, let's say $[a, b]$ closed, a closed interval, then as a metric space with the distance of the absolute value, this is a complete metric space. Because when you take a Cauchy sequence, a Cauchy sequence converges in $\mathbb{R}$, so the limit will belong to $\mathbb{R}$, but then since this is a closed interval, the limit will belong to your space. Right? More interesting example: $\mathbb{Q} \subset \mathbb{R}$. So, $A = \mathbb{Q}$. If the subset is $\mathbb{Q}$, the rational numbers, the rational numbers are not complete. That's because you can take a sequence of rational numbers converging to $\sqrt{2}$, right?

Examples: Closed Intervals and Rational Numbers

Closed Intervals: The space $A = [a, b]$ with the Euclidean metric is **complete**. Any Cauchy sequence in $[a, b]$ converges to a limit in $\mathbb{R}$, and since the set is closed, that limit must lie within $[a, b]$.

Rational Numbers: The space $(\mathbb{Q}, d)$ is **not complete**.

- There exist sequences $(x_n) \subset \mathbb{Q}$ such that $x_n \to \sqrt{2}$ in $\mathbb{R}$.
- Such a sequence is Cauchy in $\mathbb{Q}$ (because it converges in the larger space $\mathbb{R}$), but its limit $\sqrt{2}$ is not in $\mathbb{Q}$.

Explanation of Steps: The incompleteness of $\mathbb{Q}$ was the primary historical motivation for the construction of the real numbers $\mathbb{R}$. We *"fill the holes"* in $\mathbb{Q}$ to ensure that every sequence that *"should"* converge actually has a limit point.

Spoken Protocol [00:07:34 - 00:09:25]

You can take some sequence of rational numbers that as real numbers converge to $\sqrt{2}$. So, if you were — if your space were the rational numbers, the — so the fraction, right? Just fractions of integers. And you didn't invent yet the — the reals, you would have a problem because you try — so you don't have this completeness property. And this is the whole point of inventing the reals, right? To make the rationals complete.

Transition

The lecturer erases the right chalkboard and prepares to state the main theorem of the section.

Completeness of $\mathbb{R}^n$

Spoken Protocol [00:09:25 - 00:11:08]

Okay, so now I can do two things. So, I don't see more questions, right? So, we can — we can start to introduce open and closed sets in a metric space. Okay? And then — the last five minutes I will — I will discuss a bit because this year you already saw with Professor Figalli the construction of the reals, uh, with these Dedekind cuts, I think, right? So, you have a construction of the real numbers, but one can also construct really as — as I said here, as a completion. So, if you make $\mathbb{Q}$ complete, there is a — a way, an abstract way to make any metric space complete. We can discuss this very quickly for those interested. And this procedure applies to $\mathbb{Q}$ and gives you $\mathbb{R}$. And this is another way to construct a model of $\mathbb{R}$. But since this is not strictly needed now, it's just for extra information, and I will rush and be quick on that, and I will spend just the last five minutes. Now let's advance and introduce open and closed sets. Ah, okay, or maybe I can — no, I think that — sorry, maybe now I can prove, it's better that I prove that $\mathbb{R}^n$ is complete. It's a good time.

Theorem: Completeness of $\mathbb{R}^n$

Theorem 9.33 (Completeness of $\mathbb{R}^n$). *The space $\mathbb{R}^n$ equipped with the Euclidean distance d_{Eucl} is a complete metric space.*

Spoken Protocol [00:11:08 - 00:12:38]

So, let's say this theorem. *[Writes on the board]* Sorry. $\mathbb{R}^n$ with the Euclidean distance is complete. So, this is a theorem you have for $\mathbb{R}$. And actually, to prove the theorem for $\mathbb{R}^n$, I will use the result in $\mathbb{R}$. I will use that $\mathbb{R}$ is complete, that this is a highly non-trivial thing. It's almost axiomatic. You define — you define at the beginning $\mathbb{R}$ axiomatically to make it complete. So, this is hard. But once I know that $\mathbb{R}$ is complete, it's very easy to show that $\mathbb{R}^n$ is complete. Right? So, I will prove this theorem, but to prove this theorem I will need some lemma that will be useful for you many, many times, that says the following.

Coordinate-wise Convergence

Spoken Protocol [00:12:38 - 00:14:12]

So, if x_n... ah, now we will have the usual notational problem. So, I will put x_m *[Writes on the board]* a sequence in $\mathbb{R}^n$. I did not use sub-index n here, because when I'm in $\mathbb{R}^n$, n will be in this case the dimension. So, I cannot use n here, okay? And then, when we have a sequence of points like this, x_m, remember that for a point in $\mathbb{R}^n$ — so this is a bit of notation, but it's important that we clarify right now. So, in $\mathbb{R}^n$, we were denoting the points $x = (x_1, \ldots, x_n)$. And these are the coordinates of the point that are n real numbers.

Notation for Sequences in $\mathbb{R}^n$

To avoid conflict with the dimension n, we denote a sequence in $\mathbb{R}^n$ using the index m:

$$(x_m)_{m \geq 0} \subset \mathbb{R}^n$$

Each element x_m of the sequence is a vector with n coordinates:

$$x_m = (x_{m,1}, x_{m,2}, \ldots, x_{m,n})$$

where $x_{m,i} \in \mathbb{R}$ represents the i-th coordinate of the m-th vector in the sequence.

Spoken Protocol [00:14:12 - 00:15:50]

But now, if I have a sequence of points, we will denote the coordinates like this: x_m is a point in the sequence, and we will denote the coordinates $x_{m,1}, x_{m,2}, \ldots, x_{m,n}$. *[Writes on the board]* You see, there is one sub-index that is which element of the sequence, and the second sub-index which coordinate. Right? It must be like that. So, this is the notation. And if you want, there is a small notational conflict because when you see x_m, you are not sure — unless you — well, you will be sure by the context. But if I only tell you x_m, what is this? Is the m-th coordinate of the point x, or the m-th element of a sequence?

You will know always looking at the context. But this is a small notational conflict that I have pointed out in the notes also.

Lemma: Coordinate-wise Convergence

Lemma 9.26 (Convergence: in $\mathbb{R}^n$ versus coordinate-wise)**.** *Let $(x_m)_{m\geq 0}$ be a sequence in $\mathbb{R}^n$. Then x_m converges to x if and only if every coordinate converges to the corresponding coordinate of the limit:*

$$x_m \to x \iff x_{m,i} \to x_i \quad \text{for all } i = 1, \dots, n$$

Spoken Protocol [00:15:51 - 00:17:20]

So, the way to find if a sequence converges in $\mathbb{R}^n$, it's just look at each coordinate, and if each coordinate is converging to the corresponding coordinate of the limit — these are the coordinates of that, right? — then you have the convergence, and this is if and only if. And we will first prove the lemma, and from this lemma we will get immediately that the $\mathbb{R}^n$ is complete. Okay.

Spoken Protocol [00:17:21 - 00:18:42]

Okay, let's do this implication, right? The implication to the right ($\implies$). So, I start — starting point is I know that x_m converges to x. What does it mean? *[Writes on the board]* For all ϵ positive, exists capital N such that the modulus of $x_m - x$ is less than ϵ if m is bigger than N. This is the definition of convergence in a metric space, and I'm using that in my metric space the distance is the modulus of the difference of points. So, this is the same as the distance between x_m and x. It's the definition of the Euclidean distance. Right?

Proof: Implication $\implies$

Assume $x_m \to x$. By definition:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m \geq N, \|x_m - x\| < \epsilon$$

where $\|\cdot\|$ denotes the standard Euclidean norm.

Spoken Protocol [00:18:42 - 00:20:12]

Now I want to prove that this implies that each coordinate converges. But you see that — *[Writes on the board]* this thing implies that $|x_{m,i} - x_i|$ in absolute value... what is this? This is less than the sum for $j = 1$ to n of the root... no, the root goes outside. Of $(x_{m,j} - x_j)^2$. Right? If I take — so given some i, so given i in the set $\{1, \dots, n\}$, so given some coordinate i, $|x_{m,i} - x_i|$ is less than the sum for all coordinates of the squares and then you take the square root. Right? But this is the norm by definition. And I'm saying that this eventually — so this will be less than ϵ. Right?

Coordinate Bound by Norm

For any fixed coordinate $i \in \{1, \ldots, n\}$, we have the fundamental inequality:

$$|x_{m,i} - x_i| = \sqrt{(x_{m,i} - x_i)^2} \leq \sqrt{\sum_{j=1}^{n} (x_{m,j} - x_j)^2} = \|x_m - x\| \tag{9.1}$$

Explanation of Steps: The absolute difference of a single coordinate is always less than or equal to the total Euclidean distance between the vectors. Since $\|x_m - x\| < \epsilon$ for $m \geq N$, it follows immediately that $|x_{m,i} - x_i| < \epsilon$ for the same $m \geq N$. This proves that $x_{m,i} \to x_i$ for each i.

Spoken Protocol [00:20:12 - 00:21:30]

And this is what I wanted to show, right? Given ϵ, exists some N — here is the same capital N — such that when m is bigger than N, the coordinate — the element $x_{m,i} - x_i$ will be less than ϵ. This just follows by this inequality, the comparison between the difference of coordinates and the square root. So, this implication is rather easy and is done here. *[Draws a Q.E.D. box for the first part]*

Spoken Protocol [00:21:30 - 00:22:55]

And the other implication ($\Longleftarrow$) is almost the same but is slightly harder. Because now I start at the other point. I know suppose that for each coordinate i, $x_{m,i}$ converges to x_i. So, I'm putting the definition. *[Writes on the board]* This means that for all i, given i, for all ϵ exists some N that depends on i, that's why I call it N_i, such that this holds. Good. And now, if I can do it this, I can do it — I can take ϵ divided by n, so I can take ϵ/n, right? It's just you are taking a different ϵ. This is true.

Proof: Implication $\Longleftarrow$

Assume $x_{m,i} \to x_i$ for all $i \in \{1, \ldots, n\}$. By definition, for each i:

$$\forall \epsilon' > 0, \exists N_i \in \mathbb{N} \text{ s.t. } \forall m \geq N_i, |x_{m,i} - x_i| < \epsilon'$$

To prove vector convergence, we choose $\epsilon' = \frac{\epsilon}{\sqrt{n}}$.

Spoken Protocol [00:22:55 - 00:24:45]

And then what you do now is that, okay, if this happens, then the modulus of $x_m - x$... what is this? This is the sum for $i = 1$ to capital n of $(x_{m,i} - x_i)^2$ square root. Now I'm saying that for if m I will take m bigger than the maximum for i between 1 and n of these numbers N_i. I take the maximum of all these numbers, I call this N. If I take this N that is the maximum of all these numbers and m is larger, it would satisfy all the time at the same time all these conditions. And then I will get this is less than the root of n times, and then this is ϵ^2 over n^2. Each is less than ϵ/n, so when I square is ϵ^2/n^2. And this is less than ϵ divided by root n, that is less than ϵ.

Vector Convergence from Coordinate Convergence

Let $N = \max(N_1, N_2, \dots, N_n)$. Then for all $m \geq N$, the condition $|x_{m,i} - x_i| < \frac{\epsilon}{\sqrt{n}}$ holds for all i simultaneously. Substituting this into the Euclidean norm:

$$\begin{aligned} \|x_m - x\| &= \sqrt{\sum_{i=1}^{n} (x_{m,i} - x_i)^2} \\ &< \sqrt{\sum_{i=1}^{n} \left(\frac{\epsilon}{\sqrt{n}}\right)^2} \\ &= \sqrt{\sum_{i=1}^{n} \frac{\epsilon^2}{n}} \\ &= \sqrt{n \cdot \frac{\epsilon^2}{n}} = \sqrt{\epsilon^2} = \epsilon \end{aligned}$$

Explanation of Steps: By choosing a sufficiently large N such that every coordinate is *"close enough"* to its limit, we ensure that the total Euclidean distance (the sum of the squared differences) is also small. This establishes that $x_m \to x$ in $\mathbb{R}^n$.

Spoken Protocol [00:24:45 - 00:25:20]

You see, this establishes that implication. So, I assumed that the coordinates converge, I needed to take a smaller ϵ divided by n, but then the vector converges. Right?

Spoken Protocol [00:25:20 - 00:26:35]

And now, with exactly the same trick, exactly the same setup, I can prove the theorem. Proof of the theorem, today's theorem that is the completeness of $\mathbb{R}^n$. $\mathbb{R}^n$ complete. Right? So, how do you prove that something is complete? You need to take — you give me a Cauchy sequence and I must show that the Cauchy sequence converges.

Proof of Theorem 9.33: Completeness of $\mathbb{R}^n$

Let $(x_m)_{m \geq 0}$ be a Cauchy sequence in $\mathbb{R}^n$.

Step 1: Coordinate-wise Cauchy: By the inequality in Equation 9.1, for each $i \in \{1, \dots, n\}$:

$$|x_{m,i} - x_{k,i}| \leq \|x_m - x_k\|$$

Since (x_m) is Cauchy in $\mathbb{R}^n$, it follows that each coordinate sequence $(x_{m,i})_{m \geq 0}$ is Cauchy in $\mathbb{R}$.

Step 2: Completeness of $\mathbb{R}$: Since $\mathbb{R}$ is complete, each coordinate sequence converges to some $x_i \in \mathbb{R}$:

$$\lim_{m \to \infty} x_{m,i} = x_i$$

Step 3: Vector Convergence: Define the vector $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. By Lemma 9.26, since each coordinate converges, the vector sequence converges:

$$\lim_{m \to \infty} x_m = x$$

Thus, every Cauchy sequence in $\mathbb{R}^n$ converges, proving that $\mathbb{R}^n$ is complete. Q.E.D.

Spoken Protocol [00:26:35 - 00:28:08]

So, in one sentence because it's time, the proof of the completeness of $\mathbb{R}^n$ is just pass to the components, look at the components, the components are Cauchy sequences, they converge, and then you put together the components to make the point and prove the existence of the limit. Right? I'm using the lemma, this implication.

Student Interaction

Could you please elaborate why you use the minimum with the distances there instead of, for instance, the maximum?

Spoken Protocol [00:28:12 - 00:28:47]

[*The lecturer looks at the board*] Uh... it's the maximum. Okay. Thank you. [*Corrects the board*] Thank you. [*The lecturer begins to pack up his equipment as the class ends*]

End of Lecture

The lecturer corrects a minor notation on the board and concludes the session. The students applaud as he prepares to leave.

Friday: February 20th 2026

Topology of Metric Spaces: Open and Closed Sets

Context

This lecture marks the transition from the basic definitions of metric spaces to the study of their **topology**. The lecturer, Joaquim Serra, reviews the concepts of convergence and completeness before introducing the fundamental building blocks of topological spaces: open and closed sets. The discussion emphasizes the use of axioms to prove properties that will later be applied to multivariable differentiation and integration.

10.1 Review of Previous Lectures

Spoken Protocol [00:00:00 - 00:01:51]

We can continue. Uh, good afternoon. We can — we can continue where we left it last time. So, I remind — so just a quick summary of what we saw the last lectures. First lecture was mostly introduction, but then we saw metric spaces. And I told you — because people already asked me, even though I thought the first day — why we studied metric spaces. So, the model — whenever it gets too abstract, the model you need to have in mind all the time for this (X, d) metric space is $\mathbb{R}^n$ with the Euclidean distance, okay? So, whenever you have doubts, just plug $\mathbb{R}^n$ where there is an X (a capital X) and d is the usual — the modulus of the difference of the points. And, um, I said that I wanted to do this in this more general framework because later, in the last part of the course, we will use other spaces. Okay?

Summary of Metric Space Foundations

The core structure studied thus far is the metric space (X, d).

- **Standard Model:** $(\mathbb{R}^n, d_{\text{Eucl}})$, where $d(x, y) = \|x - y\|$.
- **Sequences:** $(x_n)_{n\geq 0} \subset X$ converges to $x \in X$ ($x_n \to x$) if $d(x_n, x) \to 0$.
- **Completeness:** A space is **complete** if every Cauchy sequence converges. We proved that $\mathbb{R}^n$ is a complete metric space.

Spoken Protocol [00:01:51 - 00:02:22]

And for good reason. It's not — I mean, we will use to prove some theorems in the —

in the course, the space of continuous functions. Uh, but — if you want for the — this is not something that belongs to mathematicians, right? Let me say very quickly for the physicists: so in classical mechanics, the points in the space is a vector in $\mathbb{R}^n$, right? But when you — you will go to quantum mechanics, what is a point? What is the position of a particle in space? Is the wave function. This is a function. So, the function — the wave functions are naturally points in your space, right? So, this is what we are saying here. We are trying to say things more general to be able to understand, um, the things that come naturally in the physics, for instance.

Insight: **The Physical Motivation for Abstraction**

The lecturer provides a profound justification for studying abstract metric spaces. In classical mechanics, a "point" is a coordinate in $\mathbb{R}^n$. In quantum mechanics, a "state" is a wave function. By treating functions as points in a metric space, the tools of Analysis (limits, continuity, topology) become directly applicable to the most advanced models of physics.

Spoken Protocol [00:02:22 - 00:03:13]

Okay. So, now, in this framework of metric spaces, we saw what it means a sequence, what it means that a sequence converges, what is a Cauchy sequence, what is a complete metric space, and we proved that $\mathbb{R}^n$ is always a complete metric space using that $\mathbb{R}$ is complete. This last time. And today we will see, uh, what is the definition of open and closed sets in a metric space and the notion of continuous functions in the setup of metric spaces. So, let me start with the definition of open set. So, let me start maybe with the definition of open ball. Open ball.

10.2 Topology of Metric Spaces

10.2.1 Open and Closed Sets

Definition: Open Ball

Definition 10.38 (Open Ball)**.** We define the **open ball** of radius $r > 0$ centered at a point $x \in X$ in a metric space (X, d) as:

$$B(x,r) = \{y \in X \mid d(y,x) < r\}$$

Explanation of Steps: The open ball is the set of all points whose distance to the center x is strictly less than the radius r. The strict inequality ($<$) is what makes the ball *"open"*—it does not include its boundary.

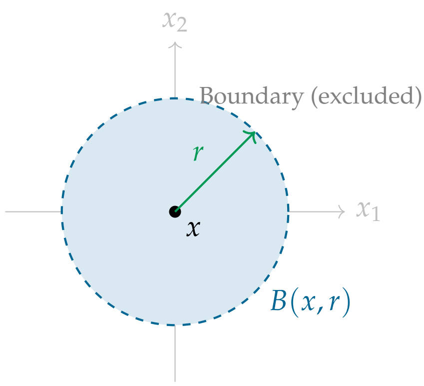
Notation Note: The lecturer notes that $B_r(x)$ is an equivalent and common notation for $B(x,r)$ found in many textbooks.

Spoken Protocol [00:03:13 - 00:05:12]

So, we — we define — define the open ball, open ball of radius r positive centered at a

given point x in my metric space as — and we will denote with this: is ball centered at x with radius r. Sometimes you will see equivalent notation is the ball of radius r centered at x. This is also in many notes and books. Is the same. Okay? This is by definition the set of points in my space such that the distance — sorry, the points will be y — such that the distance between y and x is less than r, right?
So, if you want a drawing of that, in the Euclidean space — in a — in a general metric space I cannot draw anything because it's too general — but in the Euclidean space, let's say in $\mathbb{R}^2$, the metric ball would be the ball. So, the — the open ball would be this, right? *[Draws a circle on the board]* What is inside of the — of the ball. And it's open because you don't include — if you want I can — I can put the boundary of the ball, that we will introduce in a — today also in a — in a general topological space what the boundary means. This surface of the ball is not included, because there is a strict inequality, okay? The points that are at exactly distance r, I don't take them. This is the open ball.

Visualizing the Open Ball in $\mathbb{R}^2$



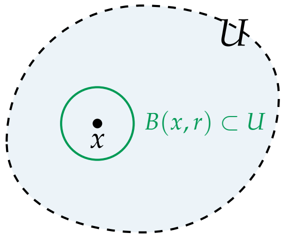
Spoken Protocol [00:05:12 - 00:06:15]

Now, next definition is open and closed sets. So, we say that a set U of my metric space X is open if for every x in U, exists some positive radius r (depending on x, possibly) such that the ball of radius r centered at x is contained in U. So, I will draw some in $\mathbb{R}^2$ some open set. Is essentially any — any set — this is a bit — heuristically, this is the idea, right? Then we will see concrete open sets. But is a set that does not include its contour. So, its boundary. Again, I — I didn't define the boundary yet, but now you know — you — you have some common sense knowledge of what is a boundary of this set, that is this part — is the skin of the set, right? If the set is the inside without the skin, this is open.

Definition: Open and Closed Sets

Definition 10.39 (Open Set). A subset $U \subseteq X$ of a metric space (X, d) is **open** if every point in U is the center of some open ball entirely contained within U:

$$\forall x \in U, \exists r > 0 \text{ s.t. } B(x, r) \subseteq U$$



Explanation of Steps: The intuition behind an open set is that no point in the set is on the *"edge"*. For any point x, you can always move a tiny distance in any direction and still remain inside the set.

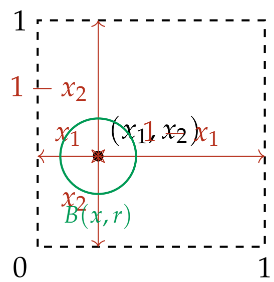
Spoken Protocol [00:06:15 - 00:08:37]

Why? Because any point here, you can take a ball — depending on the distance to the boundary, right? Depending on the distance to the skin. You make the ball super small and it will be contained in the set. Is — as I said, any point in the interior will be at positive distance from the skin.
So, if you want a more concrete example, take the set $(0,1) \times (0,1)$, the square, open square, subset $\mathbb{R}^2$. This — prove that it is open. Exercise. So, you take this square... *[Draws a dashed square on the board]* You take a point here. A point in the square will have two coordinates. And now, what is the definition of open? If you take the point — this point is (x_1, x_2) — you need to find a — you need to construct a radius r positive such that the whole ball centered at this given point belongs to the — to the square. In the case of the square, what is the formula for the radius? Well, you can take r equal, I don't know, something like the minimum between this distance, this distance, this distance, and this distance. So, it will be something like the minimum between $x_1, x_2, 1 - x_1$ and $1 - x_2$. You can check that this radius works. The minimum of these four numbers. Right?

Example: The Open Unit Square

Consider the open unit square in $\mathbb{R}^2$:

$$U = (0,1) \times (0,1) \subset \mathbb{R}^2$$



Explanation of Steps: To prove U is open, for any $(x_1, x_2) \in U$, we must find $r > 0$ such that $B(x,r) \subset U$. The distance to the nearest boundary is given by:

$$r = \min(x_1, x_2, 1 - x_1, 1 - x_2)$$

Since $0 < x_1, x_2 < 1$, all four terms are strictly positive, so $r > 0$. Any point y within distance r of x will necessarily have coordinates between 0 and 1, remaining inside the square.

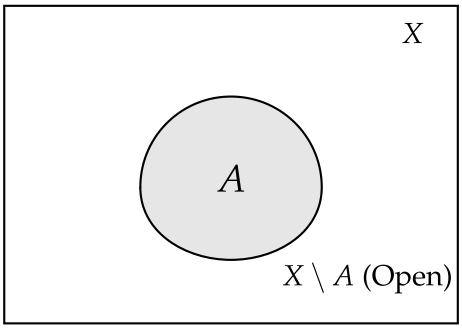
Spoken Protocol [00:08:37 - 00:10:16]

Okay. Second point in the definition is what is a closed set. So, A subset X is called closed if the complement is open. $X - A$ is open. This set $(0,1) \times (0,1)$ is not closed. You can show. Instead, this set — for instance, $[0,1] \times [0,1]$ — this set is closed. Now it needs to contains the skin to be closed. Because then, when you pass to the open — to the complement that is by definition open, it need not to contain the boundary, right? So, the closed sets, we draw them with their skin. This is closed. A.

Definition: Closed Set

Definition 10.40 (Closed Set)**.** A subset $A \subseteq X$ is **closed** if its complement in X is an open set:

$$A \text{ is closed} \iff X \setminus A \text{ is open}$$



Explanation of Steps: A closed set *"contains its boundary"*. For example, the closed unit square $[0,1] \times [0,1]$ is closed because its complement (everything outside the square) is an open set.

Spoken Protocol [00:10:16 - 00:12:03]

Okay. And maybe the last part of this definition that you will not use — we will not use actually — but — but is something that is a — a common term that you will find in books and it's good that you know, is what is the topology. So, you will see in the notes you will see the word topology many times. You are not yet studying topology but — just for information, the word topology — topology generated — so the — the topology associated to the distance d is the — we can put a $\mathcal{T}$ like this, a sort of calligraphic $\mathcal{T}$, is the collection of all the open sets. This is the topology. So, when we — when we say that something is a topological notion, it has to do with this collection of open sets.

Definition: Topology

Definition 10.41 (Topology)**.** The **topology** $\mathcal{T}$ on a metric space (X, d) is the collection of all open subsets of X:

$$\mathcal{T} = \{U \subseteq X \mid U \text{ is open}\}$$

Spoken Protocol [00:12:03 - 00:13:41]

Now, to see a bit more what are — because this is still a bit abstract — to see a bit more practically how we identify open and closed sets, uh, we are going to do it through sequences. Okay? So, this is the... well, let me first do a lemma, maybe, that it we will use many times. Lemma: unions and intersections of open sets. This is one of the basic properties. It's — it's very simple, but since we will use many, many times, I prove it now. So, it says that — I will put it with words and then we will prove with symbols. So, arbitrary unions of open sets are open.

Proposition: Unions and Intersections of Open Sets

Proposition 10.43. In any metric space (X, d):

(a) The union of an **arbitrary** collection of open sets is open.

(b) The intersection of a **finite** collection of open sets is open.

Spoken Protocol [00:13:41 - 00:15:50]

So, this means that if you have some collection $\{U_i\}_{i\in I}$ of U_i subset X open, then the union of this collection (that can be — the union of the collection is also open). Right? And also finite intersections of open sets are open. Right? So, this means that if I is finite (is a finite set), the intersection — let's say, let's put it like this: the intersection of U_i for i between 1 and capital N (a bounded number) — this is open set.

Spoken Protocol [continued]

The proof of this is just using the definition. So, if I want to check that a union is open, I take a point x in this set that is the union. Let me call this in the union. And then this in particular x belongs to one of them, because it's a union. I call i_0 is one of the is. I any of them. And then since this — this guy is open, exists some radius such that the ball of radius r centered at x is contained in U_{i_0}. But this is contained in the union. Which proves that this union is open, right?

Proof: Arbitrary Unions

Let $\{U_i\}_{i\in I}$ be a collection of open sets. Let $x \in \bigcup_{i\in I} U_i$.
Step 1: By the definition of union, there exists some index $i_0 \in I$ such that $x \in U_{i_0}$.
Step 2: Since U_{i_0} is open, there exists $r > 0$ such that $B(x,r) \subseteq U_{i_0}$.
Step 3: Since $U_{i_0} \subseteq \bigcup_{i\in I} U_i$, it follows that $B(x,r) \subseteq \bigcup_{i\in I} U_i$.
Thus, the union is open.

Spoken Protocol [continued]

With the finite intersection, you can do as an exercise. But I will tell you how to do, okay? Now explain the proof with, yeah. [*The lecturer retrieves the catch-box microphone*]

Student Question

Except for the empty set and the set X itself, can you give us some examples of both closed and open sets?

Spoken Protocol [continued]

Um, we will discuss this when we — when we discuss connected spaces. Okay? This is a good question. So, the — the whole space and the empty set are always open and closed, and I will emphasize this when we see connected spaces. So, you need to wait a bit more. I could give an example, but it's not very interesting. It's just a disconnected space like the — the union, the disjoint union of two intervals or something like that, then each interval is open and closed. But this will not be very interesting. You will see the interest of this question when we see connected spaces. Okay? But — is a fair question.

Proof: Finite Intersections

Let $U_1, \ldots, U_N$ be open sets. Let $x \in \bigcap_{i=1}^N U_i$.
Step 1: Since x is in the intersection, $x \in U_i$ for all $i \in \{1, \ldots, N\}$.
Step 2: Since each U_i is open, for each i there exists $r_i > 0$ such that $B(x, r_i) \subseteq U_i$.
Step 3: Let $r = \min(r_1, r_2, \ldots, r_N)$. Since we are taking the minimum of a **finite** number of positive values, $r > 0$.
Step 4: For this r, $B(x, r) \subseteq B(x, r_i) \subseteq U_i$ for all i. Therefore, $B(x, r) \subseteq \bigcap_{i=1}^N U_i$.
Thus, the finite intersection is open.

Spoken Protocol [00:18:05 - 00:19:05]

So, I am saying that when — when you take open sets, when you take finitely many intersections of open sets, uh, you get something open. Why? I will say with words, and then you try to write down, okay? So, you — you have finitely many open sets. And now you take a point that is in the intersection of these finitely many. Right? This means that the point is in all of them. And now, for each of the sets, you pick a radius that is different in every time, but there are finitely many radii that you can put your ball in the open set. And if you take the minimum of all these radii, it will be good. With the minimum of the finitely many radii, you will fall inside of the intersection. Okay?

Spoken Protocol [00:19:05 - 00:21:42]

You don't need to understand what I said, please try to do the exercise. If there are problems doing this, uh, you have it in the notes. But I think this is a — is very good to try in these first proofs of the — of the theory that are very short one-line proofs, is very good opportunity to do as an exercise to see if we understood the definitions. Because is one line. So, it's not — you don't need to think too much.
Okay. Now, analogous lemma but for closed sets. Unions and intersections of closed sets. And it is the same but — the same but the opposite. So, it says that finite unions of closed sets are closed, and arbitrary intersections of closed sets are closed. And the proof is essentially — well, you can repeat the same proof or you can reason a bit more — or a

variant of the same proof — or you can reason passing to the complement. So, you can use that when you have a finite union — so remember that being closed is defined as the complement being open. So, if you pass to the complement, the complementation converts closed sets in open and open in closed, right? So, you can always pass statements about open sets to the complement and you get a statement about closed sets. If you have some property of open sets and you pass to the complements (all the statement), you get a property of closed sets. And this is what we have here. Why? Because X minus a union of sets is the same as the intersection of the complements, right?

Proposition: Unions and Intersections of Closed Sets

Proposition 10.44. In any metric space (X, d):

(a) The intersection of an **arbitrary** collection of closed sets is closed.

(b) The union of a **finite** collection of closed sets is closed.

Proof via De Morgan's Laws

The properties of closed sets follow directly from the properties of open sets using De Morgan's Laws:

$$X \setminus \left(\bigcup_{i \in I} A_i \right) = \bigcap_{i \in I} (X \setminus A_i) \tag{10.1}$$

$$X \setminus \left(\bigcap_{i \in I} A_i \right) = \bigcup_{i \in I} (X \setminus A_i) \tag{10.2}$$

Arbitrary Intersections: Let $\{A_i\}_{i \in I}$ be closed. Then each $X \setminus A_i$ is open. By Proposition 10.43, their arbitrary union $\bigcup (X \setminus A_i)$ is open. By Equation 10.2, this union is the complement of the intersection $\bigcap A_i$. Since the complement is open, the intersection is closed.

Finite Unions: Let $A_1, \ldots, A_N$ be closed. Then each $X \setminus A_i$ is open. By Proposition 10.43, their finite intersection $\bigcap (X \setminus A_i)$ is open. By Equation 10.1, this intersection is the complement of the union $\bigcup A_i$. Since the complement is open, the union is closed.

Spoken Protocol [00:21:42 - 00:24:23]

So, when you do the complement of a union, you get the intersection of the complements. And the same: the complement of a intersection is equal to the union of the complements. And then you apply the previous result. Because if these sets are closed, these sets will be open. And the intersection — so the finite intersection of open is open. Okay?

Let me give some example very — very easy that this property — when we said finite intersections or finite unions — is important that it is finite. That when you get infinite intersections of open sets, you can get a closed set that is not open. Example: finite is important. So, if you take the simplest possible — well, the simplest possible no, but one of the spaces that you are most familiar with that is $\mathbb{R}$ with the distance of the absolute value (with the Euclidean distance), and you take the sets $U_k = (-1/k, 1/k)$. And you take its intersection, then the intersection for k equals 1 to infinity of U_k — this gives you the point $\{0\}$. So, these are open intervals, and it's very easy because the intervals do not contain the end points, each of them is open. But when you do their intersection, you get just the point zero, and the point zero is not open because is — you cannot put any ball in that contains anything, right? The point zero is closed but is not open. So, these are open and this is not open! That's why it's important that the intersection is finitely many.

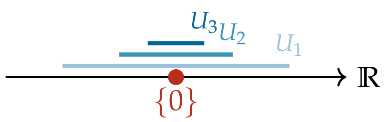
Example: Infinite Intersection of Open Sets

Example 10.45. The finiteness condition in Proposition 10.43 is necessary. Consider the collection of open intervals in $\mathbb{R}$:

$$U_k = \left(-\frac{1}{k}, \frac{1}{k}\right) \quad \text{for } k \in \mathbb{N}$$

The intersection of this infinite collection is:

$$\bigcap_{k=1}^{\infty} U_k = \{0\}$$



Explanation of Steps: While each U_k is an open set, their infinite intersection is the singleton set $\{0\}$, which is **closed** but **not open** in $\mathbb{R}$. This demonstrates that the property of openness is not necessarily preserved under infinite intersections.

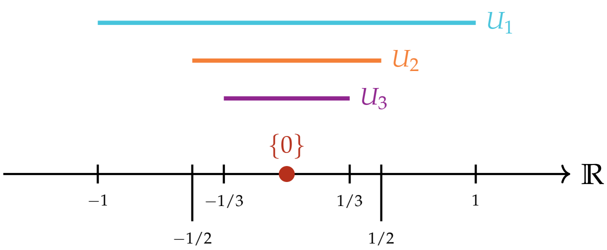
Example: Infinite Intersection of Open Sets

Example 10.45 (Infinite Intersection of Open Sets)**.** The finiteness condition for the intersection of open sets is necessary. Consider the metric space $(\mathbb{R}, d_{\text{Eucl}})$ and the following collection of open intervals:

$$U_k = \left(-\frac{1}{k}, \frac{1}{k}\right) \quad \text{for } k \in \mathbb{N} = \{1, 2, 3, \dots\}$$

Each U_k is an open set. However, their infinite intersection is:

$$\bigcap_{k=1}^{\infty} U_k = \{0\}$$



Explanation of Steps: While each U_k is an open set, as $k \to \infty$, the intervals shrink around the origin. The only point contained in **every** interval is 0. Therefore, the infinite intersection is the singleton set $\{0\}$. This set is **closed** but **not open** in $\mathbb{R}$, because any open ball $B(0,r) = (-r,r)$ around 0 will inevitably contain points other than 0. This clearly demonstrates that the property of openness is not necessarily preserved under infinite intersections.

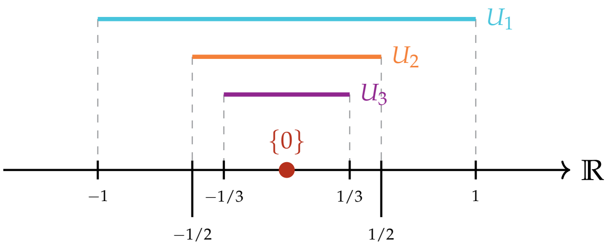
Example: Infinite Intersection of Open Sets

Example 10.45 (Infinite Intersection of Open Sets)**.** The finiteness condition for the intersection of open sets is necessary. Consider the metric space $(\mathbb{R}, d_{\text{Eucl}})$ and the following collection of open intervals:

$$U_k = \left(-\frac{1}{k}, \frac{1}{k}\right) \quad \text{for } k \in \mathbb{N} = \{1,2,3,\dots\}$$

Each U_k is an open set. However, their infinite intersection is:

$$\bigcap_{k=1}^{\infty} U_k = \{0\}$$



Explanation of Steps: While each U_k is an open set, as $k \to \infty$, the intervals shrink around the origin. The only point contained in **every** interval is 0. Therefore, the infinite intersection is the singleton set $\{0\}$. This set is **closed** but **not open** in $\mathbb{R}$, because any open ball $B(0,r) = (-r,r)$ around 0 will inevitably contain points other than 0. This clearly demonstrates that the property of openness is not necessarily preserved under infinite intersections.

Spoken Protocol [00:24:23 - 00:27:06]

Okay. So, now we will define something interesting that is what we were saying there heuristically. So, what is the interior and the boundary of a set in a metric space. So, given a set — let me call the set Ω subset X — (X,d) is a metric space. I define the interior of Ω. This is called the interior. Maybe I will not put parenthesis. I will say $\operatorname{int}\Omega$ or sometimes you put Ω with a point like this, or with a point here, depends on the source. The interior — this is what is called the interior of Ω. This is by definition the largest open set contained in Ω. So, you have your set Ω and you look at all the open sets that are contained and you get the largest one is the union of all the open sets that contain Ω. So, is the union of the U subset Ω such that U is open. Right? This is the interior.
Now, I also define the closure that you know from $\mathbb{R}$, right? The closure of some set

is the set of points x in X such that there exists some sequence x_n such that the whole sequence... *[Audio cuts off abruptly]*

Definition: Interior and Closure

Definition 10.46 (Interior)**.** The **interior** of a set $\Omega \subseteq X$, denoted $\text{int}(\Omega)$ or $\mathring{\Omega}$, is the largest open set contained within Ω. Formally, it is the union of all open sets contained in Ω:

$$\text{int}(\Omega) = \bigcup\{U \subseteq \Omega \mid U \text{ is open}\}$$

Definition 10.47 (Closure)**.** The **closure** of a set $\Omega \subseteq X$, denoted $\overline{\Omega}$, is the set of all points that can be reached as limits of sequences in Ω:

$$\overline{\Omega} = \{x \in X \mid \exists (x_n)_{n \geq 0} \subset \Omega \text{ s.t. } x_n \to x\}$$

Spoken Protocol [00:27:02 - 00:28:33]

[The lecturer is writing on the right chalkboard] Given a set — let me call the set Y (i.e., actually Ω as he writes on the board) subset X — (X, d) is a metric space. I define the interior of Ω. This is called the interior. Maybe I will not put parenthesis. I will say $\text{int}(\Omega)$ or sometimes you put Ω with a point like this *[writes $\mathring{\Omega}$]*, or with a point here, depends on the source. The interior — this is what is called the interior of Ω. This is by definition the largest open set contained in Ω. So, you have your set Ω and you look at all the open sets that are contained and you get the largest one is the union of all the open sets that contain Ω. So, is the union of the U subset Ω such that U is open. Right? This is the interior.

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$$\overline{\Omega} = \{x \in X \mid \exists (x_n)_{n \geq 0} \subset \Omega \text{ s.t. } x_n \to x\}$$

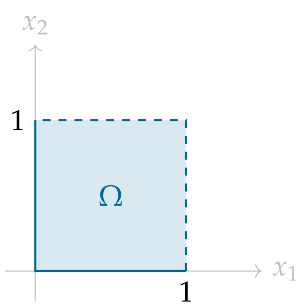
Spoken Protocol [00:28:33 - 00:30:42]

Now, I also define the closure that you know from $\mathbb{R}$, right? The closure of some set is the set of points x in X such that there exists some sequence x_n such that the whole sequence is contained in Ω and the sequence converges to x.
And here I can do a drawing, no? So, I could put some example. Let's put this example that is, say, $[0, 1) \times (0, 1)$. Closed-open, closed-open. So, how do you draw this example? It would be something like this... *[Draws a square with two solid and two dashed sides]* Right? Is the inside of this square, but you are missing this part of the boundary and this part of the boundary.

Example: Interior and Closure of a Semi-Open Square

Consider the subset $\Omega \subset \mathbb{R}^2$ defined by:

$$\Omega = [0, 1) \times (0, 1)$$



Explanation of Steps: For this set Ω:

- The **interior** $\mathring{\Omega}$ is the open square $(0,1) \times (0,1)$. It consists of all points not on any of the four edges.
- The **closure** $\overline{\Omega}$ is the closed square $[0,1] \times [0,1]$. It includes all four edges, as any point on the dashed lines can be reached by a sequence of points from the interior.

Spoken Protocol [00:30:42 - 00:32:35]

Now, can you see that these points satisfy this definition here? If I take a point here x *[points at the dashed boundary]*, here is on the boundary — so is in the closure, sorry, is on the closure. Because I can take a sequence of points from the inside, from the red part (i.e., the shaded interior), that approaches x, right? But these points that are not in my set because I am excluding them, they are also on the closure. This point here x over here is also on the closure. Because I can take a sequence from here that approaches my point.
You can also define this — and actually we will prove that — as the intersection of all the closed sets that contain my set, right? And I said this is the same. So, the closure is the smallest closed set that contains Ω.

Alternative Characterizations

The interior and closure can be characterized by their extremality properties:

- **Interior:** $\mathring{\Omega}$ is the **largest open set** contained in Ω.
- **Closure:** $\overline{\Omega}$ is the **smallest closed set** containing Ω. Formally:

$$\overline{\Omega} = \bigcap \{A \subseteq X \mid A \supseteq \Omega \text{ and } A \text{ is closed}\}$$

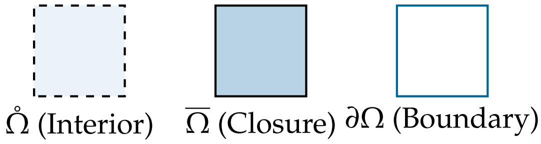
Spoken Protocol [00:32:35 - 00:34:25]

And then their difference, the difference $\overline{\Omega} \setminus \mathring{\Omega}$, this is called the boundary and is denoted with this symbol, the del symbol *[writes $\partial\Omega$]*. This is called the boundary of Ω. And this boundary of Ω is the *"skin"*. Right? And if you do any example in the plane, you will see what this is doing. So, the closure is taking all this set with the skin including the inside, and then you are removing the inside. So, when you draw the — so for this set over here, the interior I will draw the interior would be this *[draws a dashed square]*. Then when you take the closure, you get also the points that can be approximated by sequence, or you take the intersection of closed sets, you get this, the closed square. And if you want the boundary is the difference between the two, is just this part *[draws the outline of the square]* without the inside. Right?

Definition: Boundary

Definition 10.48 (Boundary)**.** The **boundary** of a set Ω, denoted $\partial\Omega$, is the set-theoretic difference between its closure and its interior:

$$\partial\Omega = \overline{\Omega} \setminus \mathring{\Omega}$$



Explanation of Steps: Heuristically, the boundary is the *"skin"* of the set. It consists of points that are *"on the edge"*—every neighborhood of a boundary point contains at least one point from Ω and at least one point from its complement $X \setminus \Omega$.

Spoken Protocol [00:34:25 - 00:35:53]

Okay. So, again all of these are perfectly common sense notions but that we can formalize and make perfectly rigorous just using these few axioms. And then I will prove maybe a lemma that actually shows that this is the same. And after this lemma you can take whatever of the two definitions, the one that is more convenient each time.
This lemma is called open and closed through sequences. How we determine if a set is open and closed using convergent sequences.

Lemma: Characterization via Sequences

Lemma 10.49 (Open and Closed Sets through Sequences)**.** *Let (X, d) be a metric space.*

(a) *A subset $U \subseteq X$ is **open** if and only if for every sequence $(x_n)_{n\geq 0}$ in X converging to a point $x \in U$, the sequence elements x_n belong to U "eventually".*

(b) *A subset $A \subseteq X$ is **closed** if and only if for every sequence $(x_n)_{n\geq 0}$ contained in A that converges to a limit $x \in X$, the limit x must also belong to A.*

Spoken Protocol [00:35:53 - 00:38:16]

So, it says that U subset X is open if and only if whenever I have a sequence x_n in my space and x_n is converging to a point — so it has a limit and the limit is a point inside the set U — then x_n belongs to U for all n up to finitely many exceptions. Right? It means, you see, I'm going to do the drawing. This is my drawing of an open set U *[draws a dashed region]*. And I have some sequence in the space here that is going to converge to this point. This sequence is going to converge to this point. It means that the sequence could start there, I don't know, but it's going to converge to this point. So, after finitely many exceptions, you will be inside of the open set. Why? Because the open set always contains a ball, and the definition of convergence means that at some point they will be inside of this ball, whatever is the radius of the ball. This is exactly the definition of convergence. And this will be the proof and I will write this, right? This is almost a proof with a picture.

Spoken Protocol [00:38:16 - 00:39:53]

So, this concept of *"up to finitely many exceptions"* is very useful and is very nice to say the thing shortly. And we give a name to this. And you have a more formal definition in the notes. So, when something happens when you have a sequence and some property — like for instance being in U — happens, is true up to finitely many exceptions, we say that this property happens *"eventually"*. So, this is a word I will use many times, okay? Whenever I have a sequence and I say that this happens eventually, means that maybe at the beginning of the sequence it's not happening, but if I wait enough for n large enough, it will happen and then it never stops to happen.

The Concept of "Eventually"

Definition 10.17 (Eventually)**.** A property $P(x_n)$ is said to hold **eventually** for a sequence $(x_n)_{n\geq 0}$ if there exists some threshold $N \in \mathbb{N}$ such that the property holds for all elements beyond that threshold:

$$\exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, P(x_n) \text{ is true}$$

Explanation of Steps: The term *"eventually"* is a rigorous shorthand for *"for all but finitely many terms"*. In the context of Lemma 10.49, it means that while the first few terms of a sequence might lie outside the open set U, the *"tail"* of the sequence must be entirely contained within U.

Spoken Protocol [00:39:53 - 00:40:14]

Yeah? *[The lecturer retrieves the catch-box microphone]* Is this *"eventually"* equivalent to saying there exists a big N such that for all n bigger than big N this holds? Exactly. Up to finitely many exceptions, it means that if you want this is equivalent by definition to eventually, which is equivalent to exists some N such that for all n bigger than N the property holds. And this N just counts how many exceptions you have. Is an upper bound for the number of exceptions. So, all of these will mean the same.

Student Question

You write the subset symbol without the equality sign at the bottom. Does it have a meaning or like that it cannot be equal or is that just notation? Like subset without the extra line at the bottom.

Spoken Protocol [00:40:14 - 00:41:02]

Ah, you mean this? *[points at $\subset$]* Is the same. You mean this? Yes. Is the same symbol. So is — well, is not the same symbol, is the same meaning. So some people use the equality since is the — I mean, since means the same I save one line, right? But many — this is very standard, many people just use this contained without the equality sign. So I will use always this contained without the equality sign.

Spoken Protocol [00:41:02 - 00:42:32]

Okay, let's do this because maybe you will find more exciting continuous functions. Since this proof over here is super similar to the previous one and is again you see that these proofs are one-line proofs, it's just using the definitions and manipulating around. So, I give this proof, I left this proof as an exercise. Try to do it again. If you understood well the definitions and you can draw a picture of that and see what's going on, this shouldn't be too difficult. But it can be a bit difficult at the beginning because there are a lot of new definitions, you need to learn them, but the best is to practice with these exercises, to try to do them. Because if I do them, it looks like a bunch of symbols that all work sometimes but you don't interiorize the definitions. The best way is to try by yourself.

Proof of Lemma 10.49 (Part 1)

Spoken Protocol [00:42:32 - 00:44:17]

Proof of number one. So, I need to do two implications. First I will do this side ($\implies$) that is the one I draw here. So, assume that U is open. And now take a point — I'm just going to put in symbols the drawing that I did here. Take x in U, any point. Then there — by definition of open set, since U is open, exists some r positive such that the ball of radius r centered at x is contained in U. Right?
And now, since by assumption x_n is converging to x, by definition of convergence what did we mean? What was the definition of convergence? Remember: given any ϵ — I called it in the definition this was the ϵ, now is r but is the same — for n large enough, bigger than a certain capital N, my sequence will belong to the ball of radius ϵ. So, since x_n converges to x, exists some capital N (depending on r) such that x_n belongs to the ball $B(x,r)$ for all n bigger than N.

Forward Implication $\implies$

Assume $U \subseteq X$ is open and let $(x_n)_{n\geq 0}$ be a sequence such that $x_n \to x \in U$.

Step 1 (Openness): Since U is open and $x \in U$, there exists a radius $r > 0$ such that:
$$B(x,r) \subseteq U$$

Step 2 (Convergence): Since $x_n \to x$, by the definition of convergence (using r as our ϵ):
$$\exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, d(x_n,x) < r$$

Step 3 (Conclusion): By the definition of the open ball, $d(x_n,x) < r \iff x_n \in B(x,r)$. Therefore:
$$\forall n \geq N, x_n \in B(x,r) \subseteq U$$

Thus, $x_n \in U$ eventually.

Spoken Protocol [00:44:17 - 00:45:53]

Now, in the definition of convergence maybe there was not this ball but there was something equivalent. There was this thing here [*points at the board*]: this is the same as saying that the distance between x_n and x is less than r. Because when a point belongs to the ball? When the distance with the center is less than r. This is if and only if. And this is what I'm saying here, end of the proof. Because I'm saying that up to finitely many exceptions that is this capital N, later on all the points of the sequence belong to the ball, and the ball is contained in U, so they belong to U. Right? So this proves this implication.

Spoken Protocol [00:45:53 - 00:48:17]

Now I will prove still of number one the second implication, the one this way ($\Longleftarrow$). And I will prove this by contraposition. Right? $A \implies B$ is equivalent to $\neg B \implies \neg A$, right? We know that. So I will prove that no left (i.e., U is not open) gives no right (i.e., there exists a sequence converging to $x \in U$ that is not eventually in U).
So now, let's do a bit of logic. What is the negation of U open? Remember now, every time: what is the definition of U open? Before computing what is the negated statement we need to remember the original statement. U open is: for every point in my set U, exists some radius such that the ball is contained in U. Right? And now we do the negation. The negation of *"for every point"* is *"exists a point"*. So the negation: U is not open is the same as exists x in U. Now it goes and it said *"for every point, exists a radius"*. Now the exists when you negate is *"for every radius"* positive, the ball is contained is *"is not contained"*. Is the ball centered at x with radius r is not contained in U.

Implication $\Longleftarrow$ via Contraposition

We prove the converse by contraposition. Assume U is **not** open.
Step 1 (Negating Openness): The negation of the definition of an open set is:

$$\exists x \in U \text{ s.t. } \forall r > 0, B(x,r) \not\subseteq U$$

Step 2 (Finding Points Outside): The condition $B(x,r) \not\subseteq U$ implies that for every r, there exists at least one point in the ball that is **not** in U:

$$\forall r > 0, \exists x_r \in B(x,r) \cap (X \setminus U)$$

Step 3 (Constructing the Sequence): We construct a sequence by choosing $r = \frac{1}{n}$ for each $n \in \mathbb{N}$. For each such r, we pick a point x_n such that:

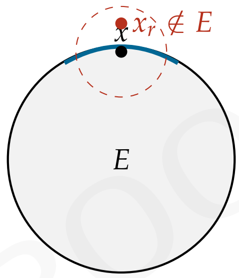
$$x_n \in B(x, 1/n) \quad \text{and} \quad x_n \notin U$$

Step 4 (Convergence): Since $x_n \in B(x, 1/n)$, we have $d(x_n, x) < \frac{1}{n}$. As $n \to \infty$, the distance $d(x_n, x) \to 0$, which means $x_n \to x$.

Spoken Protocol [00:48:17 - 00:50:14]

And that the ball is not contained in U, this is the same as saying that there exists some point, let's call it x_r because depends on r, that belongs to the ball centered at x with radius r intersected with the complement of U. What I'm saying here? Now I will do a drawing of that. So I take a set that is this one over here *[draws a region with a solid boundary segment]*, let me call it maybe E that is not open. Heuristically, the drawing of this set if the skin is not contained would be open, so I draw it like this because there is a piece of skin then is not open. Okay? This is just to make the drawing.
Now, I'm saying: there exists a point that would be one of these points of the skin x such that for every radius that I can choose here, I can find a point outside my set. So there is some point in the set that can be approximated by a sequence of points from outside the set.

Visualizing the Contraposition



Spoken Protocol [00:50:14 - 00:52:32]

Taking $r = 1/n$ for n greater than or equal to 1, I produced a sequence of points — so these balls are smaller and smaller — and these points x_r — so I call x_n this point x_n is the x for $r = 1/n$. These points are in balls closer and closer to x, so they are converging to the point x. Right? So I found a sequence that is converging to the point x and that comes from the outside. But then, this is false (i.e., contradicts the assumption) because I'm saying that up to finitely many exceptions the sequence should be contained in U. And all of the elements of this sequence are outside of U, so this is false. So this contradicts that x_n belongs to U eventually, and it ends the proof of this implication. I proved that no left implies no right.

Conclusion

Contradiction: We have found a sequence (x_n) such that $x_n \to x \in U$, but $x_n \notin U$ for **all** n. This directly contradicts the property that x_n must be in U eventually.

Thus, the contrapositive is proven: if the *"eventually in U"* property holds for all convergent sequences, then U must be open.

Q.E.D.

Spoken Protocol [00:52:32 - 00:54:25]

Okay, is this okay, this type of proof? So I see this is the usual problem: when I see people that are sleeping, I don't know if they are completely lost, not sleeping but very serious, I don't know if they are completely lost, if they find it super boring or what is the problem. So if someone has questions and wants to participate, we can slow down and ask questions, otherwise we continue with other stuff. We still need to do this one (i.e., Part 2 of the Lemma), right?

Spoken Protocol [00:54:25 - 00:55:39]

Since this proof over here is super similar to the previous one and is again you see that these proofs are one-line proofs, it's just using the definitions and manipulating around. So I give this proof, I left this proof as an exercise. Try to do it again. If you understood well the definitions and you can draw a picture of that and see what's going on, this shouldn't be too difficult. But it can be a bit difficult at the beginning because there are a lot of new definitions, you need to learn them, but the best is to practice with these exercises, to try to do them.

Spoken Protocol [00:55:39 - 00:57:02]

Maybe I can put another exercise before continuity. You have more in the exercise list. Prove — now that you have all of these, you can prove that if (X, d) is a complete metric space and A is a subset of X that is closed, then A with the same distance restricted is complete. Example: we know that $\mathbb{R}^n$ is a complete metric space. Then any closed subset of $\mathbb{R}^n$ will be a complete metric space when you restrict the distance. And now you have all the ingredients to prove that. It's not too difficult, is a good exercise.

Exercise: Completeness of Closed Subspaces

Exercise 10.50. Let (X, d) be a complete metric space. Prove that a subset $A \subseteq X$ is complete (as a metric space in its own right) if and only if A is closed in X.

End of Segment

The lecturer concludes the discussion on the topological properties of sequences and prepares to transition to the next topic: Continuity.

CHAPTER 11

Monday: February 23rd 2026

Topology of Metric Spaces

Context

This lecture transitions from the basic definitions of metric spaces and sequences to the study of **Topology**. The lecturer, Joaquim Serra, introduces the fundamental concepts of open and closed sets, interior, and closure. These concepts provide the language necessary to discuss continuity and limits in a more abstract and powerful framework, moving beyond the specific Euclidean structure of $\mathbb{R}^n$.

Initial Board State

The left chalkboard contains a summary of previous lectures:

- (X, d) metric space $[\mathbb{R}^n, d_{\text{Eucl}}]$
- $(x_n)_{n\geq 0} \quad x_n \to x$
- Cauchy seq. | Complete metric space

Below this, the plan for today is written:

- Today: Open & closed sets
- Continuous functions

11.1 Recall and Motivation

Spoken Protocol [00:00:00 - 00:01:50]

We can continue. Uh, good afternoon. We can — we can continue where we left it last time. So, I remind — so just a quick summary of what we saw the last lectures. First lecture was mostly introduction, but then we saw metric spaces. And I told you — because people already asked me, even though I thought the first day — why we studied metric spaces. So, the model — whenever it gets too abstract, the model you need to have in mind all the time for this (X, d) metric space is $\mathbb{R}^n$ with the Euclidean distance, okay? So, whenever you have doubts, just plug $\mathbb{R}^n$ where there is an X (a capital X) and d is the usual — the modulus of the difference of the points.
And, uh, I said that I wanted to do this in this more general framework because later, in the last part of the course, we will use other spaces, okay? And for good reason. It's not — I mean, we will use to prove some theorems in the course, the space of continuous functions. But, uh, if you want, for the — this is not something that belongs

to mathematicians, right? Let me say very quickly for the physicists: so, in classical mechanics, the points in the space is a vector in $\mathbb{R}^n$, right? But when you will go to quantum mechanics, what is a point? What is the position of a particle in space? Is the wave function. This is a function. So, the functions, the wave functions, are naturally points in your space, right? So, this is what we are saying here. We are trying to say things more general to be able to understand, um, the things that come naturally in the physics, for instance.

Spoken Protocol [00:01:50 - 00:02:22]

Okay. So, now, in this framework of metric spaces, we saw what it means a sequence, what it means that a sequence converges, what is a Cauchy sequence, what is a complete metric space, and we proved that $\mathbb{R}^n$ is always a complete metric space using that $\mathbb{R}$ is complete, this last time. And today we will see the definition of open and closed sets in a metric space and the notion of continuous functions in the setup of metric spaces. So, let me start with the definition of open ball.

11.2 Open Balls and Open Sets

Definition: Open Ball

Definition 11.38 (Open Ball)**.** Let (X, d) be a metric space. We define the **open ball** of radius $r > 0$ centered at a point $x \in X$ as the set:

$$B(x, r) = \{y \in X \mid d(y, x) < r\}$$

Explanation of Steps: The open ball consists of all points in the space whose distance to the center x is strictly less than the radius r. The strict inequality is what makes the ball *"open"*—it does not include its boundary.

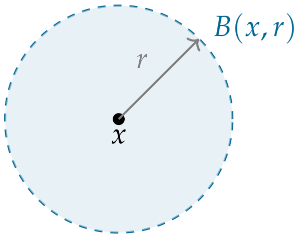
Notation

The lecturer notes that an equivalent notation often found in literature is $B_r(x)$.

Spoken Protocol [00:03:54 - 00:04:51]

So, if you want a drawing of that, in the Euclidean space — in a general metric space I cannot draw anything because it's too general, but in the Euclidean space, let's say in $\mathbb{R}^2$, the metric ball would be the ball. So, the — the open ball would be this, right? *[Draws a circle on the board]* What is inside of the — of the ball. And it's open because you don't include — if you want I can — I can put the boundary of the ball, that we will introduce today also in a general topological space what the boundary means. This surface of the ball is not included, because there is a strict inequality. The points that are at exactly distance r, I don't take them. This is the open ball.

Visualizing the Open Ball in $\mathbb{R}^2$



Spoken Protocol [00:04:51 - 00:05:52]

Now, next definition is open and closed sets. *[Writes on the board]* So, we say that a set U of my metric space X is open if for every x in U, exists some positive radius r (depending on x, possibly) such that the ball of radius r centered at x is contained in U.

Definition: Open and Closed Sets

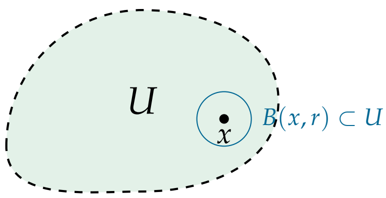
Definition 11.39 (Open and Closed Sets)**.** Let (X, d) be a metric space.

- A subset $U \subseteq X$ is **open** if every point in U is the center of some open ball entirely contained within U:
$$\forall x \in U, \exists r > 0 \text{ s.t. } B(x,r) \subseteq U$$
- A subset $A \subseteq X$ is **closed** if its complement is open:
$$A \text{ is closed} \iff X \setminus A \text{ is open}$$

Spoken Protocol [00:05:52 - 00:07:10]

So, I will draw some in $\mathbb{R}^2$ some open set. Is essentially any set — this is a bit heuristically, this is the idea, right? Then we will see concrete open sets. But it's a set that doesn't — does not include its contour, so its boundary. Again, I didn't define the boundary yet, but now you know — you have some common sense knowledge of what is a boundary of this set, that is this part — the skin of the set, right? If the set is the inside without the skin, this is open. Why? Because any point here, you can take a ball, depending on the distance to the boundary, right? Depending on the distance to the skin, you make the ball super small and it will be contained in the set. It's like I said: any point in the interior will be at positive distance from the skin.

Intuition for Open Sets



Explanation of Steps: The *"skin"* or boundary of an open set is not part of the set itself. This ensures that no matter how close a point x is to the edge, there is always a tiny bit of *"breathing room"* around it that remains entirely within U.

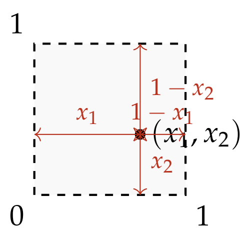
Spoken Protocol [00:07:10 - 00:08:37]

So, if you want a more concrete example, take the set $(0,1) \times (0,1)$, the square, open square, subset $\mathbb{R}^2$. This — prove that it is open. Exercise. So, you take this square... *[Draws a dashed square on the board]* You take a point here. A point in the square will have two coordinates. And now, what is the definition of open? If you take the point, this point is (x_1, x_2), you need to find a radius r positive such that the whole ball centered at this given point belongs to the square. In the case of the square, what is the formula for the radius? Well, you can take r equal, I don't know, something like the minimum between this distance, this distance, this distance, and this distance. So, it will be something like the minimum between $x_1, x_2, 1 - x_1$ and $1 - x_2$. You can check that this radius works. The minimum of these four numbers.

Example: The Open Unit Square

Consider the open unit square in $\mathbb{R}^2$:

$$U = (0,1) \times (0,1) = \{(x_1, x_2) \in \mathbb{R}^2 \mid 0 < x_1 < 1, 0 < x_2 < 1\}$$



Explanation of Steps: To prove U is open, for any $(x_1, x_2) \in U$, we choose the radius:

$$r = \min(x_1, x_2, 1 - x_1, 1 - x_2)$$

Since $0 < x_1, x_2 < 1$, all four values are strictly positive, so $r > 0$. The ball $B((x_1, x_2), r)$ is then guaranteed to be contained within the square.

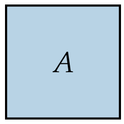
Spoken Protocol [00:08:37 - 00:10:16]

Okay, second point in the definition is what is a closed set. *[Writes on the board]* So, A subset X is called closed if the complement is open: $X \setminus A$ is open. So, now, this set — this set is not closed, you can show. Instead, this set... *[Writes $[0,1] \times [0,1]$ on the board]* for instance, this set is closed. Now it needs to contain the skin to be closed. Because then, when you pass to the complement, that is by definition open, it need not to contain the boundary. So, the closed sets, we draw them with their skin.

Example: The Closed Unit Square

The closed unit square is defined as:

$$A = [0,1] \times [0,1] = \{(x_1, x_2) \in \mathbb{R}^2 \mid 0 \le x_1 \le 1, 0 \le x_2 \le 1\}$$



Closed Set (includes boundary)

Explanation of Steps: The set A is closed because its complement, $\mathbb{R}^2 \setminus A$, is open. If you take any point outside the closed square, there is a minimum distance to the square's boundary, allowing you to fit a small open ball around that point that does not touch A.

Spoken Protocol [00:10:16 - 00:11:35]

Okay, maybe this I can erase. And maybe the last part of this definition, that you will not use — we will not use actually, but — but is something that is a common term that you will find in books and it's good that we know, is what is the topology. So, you will see in the notes you will see the word topology many times. You are not yet studying topology, but — just for information, the word topology, topology associated to the distance d is the — we can put a $\mathcal{T}$ like this, a sort of calligraphic $\mathcal{T}$, is the collection of all the open sets. This is the topology. So, when we say that something is a topological notion, it has to do with this collection of open sets.

The Topology of a Metric Space

Definition 11.40 (Topology)**.** The **topology** $\mathcal{T}$ associated with a metric space (X, d) is the collection of all open subsets of X:

$$\mathcal{T} = \{U \subseteq X \mid U \text{ is open}\}$$

11.3 Properties of Open Sets

Spoken Protocol [00:11:35 - 00:13:43]

Now, to see a bit more what are — because this is still a bit abstract — to see a bit more practically how we identify open and closed sets, we are going to do it through sequences. Well, let me first do a lemma, maybe, that we will use many times. Lemma: unions and intersections of open sets. This is one of the basic properties. It's very simple, but since we will use many, many times, I prove it now. So, it says that if you have — I will put it with words and then we will prove with symbols. So, arbitrary unions of open sets are open. This means that if you have some collection $\{U_i\}_{i \in I}$ of $U_i \subseteq X$ open, then the union of this collection (that can be the union of the collection) is also open. Right?

Proposition: Unions and Intersections of Open Sets

Proposition 11.43 (Unions and Intersections of Open Sets)**.** In any metric space (X, d):

(a) Arbitrary Unions: The union of any collection of open sets is open.

(b) Finite Intersections: The intersection of a finite number of open sets is open.

Proof of Proposition 11.43 (Part 1)

Spoken Protocol [00:13:43 - 00:15:50]

And also finite intersections of open sets are open. Right? So, this means that if I is finite, the intersection — let's say, let's put it like this: the intersection of U_i for i between 1 and capital N, a bounded number, this is open set.
The proof of this is just using the definition. So, if I want to check that a union is open, I take a point x in this set that is the union. Let me call this in the union. Right? And then, in particular, x belongs to one of them because it's a union. I call i_0 is one of the is. I fix any of them. And then, since this one is open, exists some radius r such that the ball of radius r centered at x is contained in U_{i_0}. But this is contained in the union. Which proves that this union is open, right?

Proof: Arbitrary Unions

Let $\{U_i\}_{i\in I}$ be a collection of open sets. Let $U = \bigcup_{i\in I} U_i$. Take any point $x \in U$. By the definition of union:

$$x \in U \implies \exists i_0 \in I \text{ s.t. } x \in U_{i_0}$$

Since U_{i_0} is open, by definition:

$$\exists r > 0 \text{ s.t. } B(x,r) \subseteq U_{i_0}$$

Since $U_{i_0} \subseteq \bigcup_{i\in I} U_i = U$, it follows that:

$$B(x,r) \subseteq U$$

Thus, U is open.

Q.E.D.

Student Question

Except for the empty set and the set X itself, can you give us some examples of both closed and open sets?

Spoken Protocol [00:18:16 - 00:19:15]

We will discuss this when we — when we discuss connected spaces. Okay? This is a good question. So, the — the whole space and the empty set are always open and closed, and I will emphasize this when we see connected spaces. So, you need to wait a bit more. I could give an example, but it's not very interesting. It's just a disconnected space like the union, the disjoint union of two intervals or something like that, then each interval is open and closed. But this will not be very interesting. You will see the interest of this question when we see connected spaces. Okay? But it's a fair question.

Insight: **Clopen Sets and Connectivity**

The lecturer hints at a deep topological property: in a *"connected"* space (like $\mathbb{R}^n$), the only sets that are both open and closed (often called *"clopen"*) are the empty set and the space itself. The existence of other clopen sets is the defining characteristic of a disconnected space.

11.4 Properties of Closed Sets

Spoken Protocol [00:19:15 - 00:21:15]

So, I am saying that when you take open sets, when you take finitely many intersections of open sets, you get something open. Why? I will say with words and then you try to write down, okay? So, you have finitely many open sets. And now you take a point that is in the intersection of these finitely many. This means that the point is in all of them. And now, for each of the sets, you pick a radius that is different in every time. But there are finitely many radii that you can put your ball in the open set. And if you take the minimum of all these radii, it will be good. With the minimum of the finitely many radii, you will fall inside of the intersection. Okay? You don't need to understand what I said, please try to do the exercise. If there are problems doing this, you have it in the notes. But I think this is a very good opportunity to do as an exercise to see if we understood the definitions. Because it's one line, so it's not — you don't need to think too much.

Proof Sketch of Proposition 11.43 (b): Finite Intersections of Open Sets

Proof Sketch: *Let $U = \bigcap_{i=1}^N U_i$ where each U_i is open. Take $x \in U$. Then $x \in U_i$ for all $i \in \{1, \ldots, N\}$. Since each U_i is open:*

$$\forall i, \exists r_i > 0 \text{ such that } B(x, r_i) \subseteq U_i$$

*Let $r = \min(r_1, r_2, \ldots, r_N)$. Because we are taking the minimum of a **finite** set of positive numbers, $r > 0$. Then for all i, $B(x, r) \subseteq B(x, r_i) \subseteq U_i$. Therefore, $B(x, r) \subseteq \bigcap_{i=1}^N U_i = U$, proving that U is open.* ∎

Spoken Protocol [00:21:15 - 00:24:04]

Okay. Now, analogous lemma but for closed sets. *[Writes on the board]* Unions and intersections of closed sets. And it's the same but the opposite. So, it says that finite unions of closed sets are closed and arbitrary intersections of closed sets are closed. And the proof is — well, you can repeat the same proof or you can reason a bit more — or a variant of the same proof, or you can reason passing to the complement. So, you can use that being closed is defined as the complement being open. So, if you pass to the complement, the complementation converts closed sets in open and open in closed.

So, you can always pass statements about open sets to the complement and you get a statement about closed sets. And this is what we have here. Why? Because X minus a union of sets is the same as the intersection of the complements. Right? So, when you do the complement of a union, you get the intersection of the complements. And the same, the complement of a intersection is equal to the union of the complements. And then you apply the previous result. Because if these sets are closed, these sets will be open, and the finite intersection of open is open.

Lemma: Unions and Intersections of Closed Sets

Proposition 11.44 (Unions and Intersections of Closed Sets)**.** In any metric space (X, d):

(a) Finite Unions: The union of a finite number of closed sets is closed.

(b) Arbitrary Intersections: The intersection of any collection of closed sets is closed.

Proof via De Morgan's Laws

The properties of closed sets follow directly from those of open sets using De Morgan's Laws:

$$X \setminus \left(\bigcup_{i \in I} A_i \right) = \bigcap_{i \in I} (X \setminus A_i)$$
$$X \setminus \left(\bigcap_{i \in I} A_i \right) = \bigcup_{i \in I} (X \setminus A_i)$$

Explanation of Steps: If each A_i is closed, then each complement $(X \setminus A_i)$ is open.

- For an arbitrary intersection $\bigcap A_i$, its complement is the union $\bigcup (X \setminus A_i)$. Since the arbitrary union of open sets is open, the intersection is closed.
- For a finite union $\bigcup_{i=1}^{N} A_i$, its complement is the finite intersection $\bigcap_{i=1}^{N} (X \setminus A_i)$. Since the finite intersection of open sets is open, the union is closed.

Spoken Protocol [00:24:04 - 00:26:39]

Let me give some example, very easy, that this property when we said finite intersections or finite unions is important that it is finite. That when you get infinite intersections of open sets, you can get a closed set that is not open. So, example: finite is important. So, if you take the simplest — well, one of the spaces that you are most familiar with that is $\mathbb{R}$ with the Euclidean distance, and you take the sets $U_k = (-1/k, 1/k)$. And you take its intersection, then the intersection for $k = 1$ to infinity of U_k, this gives you the point $\{0\}$. So, these are open intervals, and it's very easy because the intervals do not contain the endpoints, each of them is open. But when you do their intersection, you get just the point zero, and the point zero is not open because you cannot put any ball that contains anything. The point zero is closed but is not open. So, these are open and this is not open! That's why it's important that the intersection is finitely many. And passing to the complement you get a counterexample for the infinite unions of closed sets.

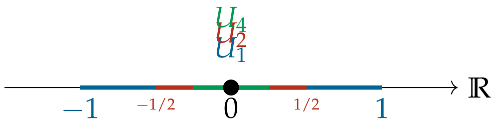
Example: Infinite Intersection of Open Sets

Example 11.45 (Infinite Intersection of Open Sets)**.** Consider the metric space $(\mathbb{R}, d_{\text{Eucl}})$. Define a sequence of open intervals:

$$U_k = \left(-\frac{1}{k}, \frac{1}{k}\right) \quad \text{for } k \in \{1, 2, 3, \dots\}$$

Each U_k is an open set. However, their infinite intersection is:

$$\bigcap_{k=1}^{\infty} U_k = \{0\}$$



Explanation of Steps: As $k \to \infty$, the intervals shrink around the origin. The only point contained in **every** interval is 0. The set $\{0\}$ is a singleton set, which is closed in $\mathbb{R}$ but not open, as any open ball $B(0, r) = (-r, r)$ contains points other than 0.

11.5 Interior and Closure

Spoken Protocol [00:26:39 - 00:28:30]

Okay, so now we will define something interesting that is what we were saying there heuristically: so what is the interior and the boundary of a set in a metric space. So, given a set Ω in my metric space X, I define the interior of Ω. This is called the interior of Ω. This is by definition the largest open set contained in Ω. So, you have your set Ω and you look at all the open sets that are contained and you get the largest one is the union of all the open sets that contain Ω. This is the interior.

Definition: Interior and Closure

Definition 11.46 (Interior and Closure)**.** Let (X, d) be a metric space and $\Omega \subseteq X$ be any subset.

- The **interior** of Ω, denoted $\text{int}(\Omega)$ or Ω°, is the largest open set contained in Ω. Formally, it is the union of all open subsets of Ω:

$$\text{int}(\Omega) = \bigcup \{U \subseteq \Omega \mid U \text{ is open}\}$$

- The **closure** of Ω, denoted $\overline{\Omega}$, is the smallest closed set containing Ω.

Spoken Protocol [00:28:30 - 00:29:59]

Now, I also define the closure, that you know from $\mathbb{R}$, the closure of some set is the set of points x in X such that there exists some sequence x_n such that the whole sequence... *[Audio cuts off abruptly]*

Sequential Characterization of Closure

A point $x \in X$ belongs to the closure $\overline{\Omega}$ if and only if it can be reached as the limit of a sequence of points from Ω:

$$\overline{\Omega} = \{x \in X \mid \exists (x_n)_{n \geq 0} \subseteq \Omega \text{ s.t. } x_n \to x\}$$

[Audio cuts off abruptly]

Spoken Protocol [00:00:00 - 00:01:32]

[The lecturer is writing the definitions of interior, closure, and boundary on the right chalkboard]

Definition: Interior, Closure, and Boundary

Definition 11.45 (Interior, Closure, and Boundary)**.** Given a set $\Omega \subseteq X$ in a metric space (X, d):

- The **interior** of Ω, denoted $\mathrm{int}(\Omega)$ or Ω°, is the largest open set contained in Ω. Formally, it is the union of all open subsets of Ω:

$$\mathrm{int}(\Omega) = \Omega^\circ = \bigcup\{U \subseteq \Omega \mid U \text{ is open}\}$$

- The **closure** of Ω, denoted $\overline{\Omega}$, is the smallest closed set containing Ω. Formally, it is the intersection of all closed sets containing Ω:

$$\overline{\Omega} = \bigcap\{A \supseteq \Omega \mid A \text{ is closed}\}$$

  Equivalently, it is the set of all points that can be reached as limits of sequences in Ω:

$$\overline{\Omega} = \{x \in X \mid \exists (x_n)_{n\geq 0} \subseteq \Omega \text{ s.t. } x_n \to x\}$$

- The **boundary** of Ω, denoted $\partial\Omega$, is the set difference between the closure and the interior:

$$\partial\Omega = \overline{\Omega} \setminus \Omega^\circ$$

Spoken Protocol [00:01:32 - 00:02:29]

Given a set $\Omega \subset X$, (X, d) is a metric space. I define the interior of Ω. This is called the interior of Ω. This is by definition the largest open set contained in Ω. So, you have your set Ω and you look at all the open sets that are contained and you get the largest one is the union of all the open sets that contain Ω (i.e., actually that are contained in Ω). This is the interior.

Spoken Protocol [00:02:29 - 00:03:43]

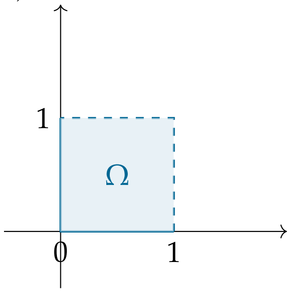
Now, I also define the closure, that you know from $\mathbb{R}$, the closure of some set is the set of points x in X such that there exists some sequence x_n such that the whole sequence is contained in Ω and then the sequence converges to x. You can also define this as the intersection of all the closed sets that are bigger than Ω (i.e., that contain Ω). So, the closure is the smallest closed set that contains Ω.

Spoken Protocol [00:03:43 - 00:05:33]

And then their difference, this minus this, this is called the boundary and is denoted with this symbol, the del symbol. This is called the boundary of Ω. And this boundary of Ω is the *"skin"*. Right? And if you do any example in the plane, you will see what this is doing. So, the closure is taking all this set with the skin, including the inside. And then you are removing the inside. So, when you draw the — for this set over here...
[Points at the square example he is about to draw]

Example: The Square $[0, 1) \times [0, 1)$

Consider the set $\Omega = [0, 1) \times [0, 1) \subset \mathbb{R}^2$.



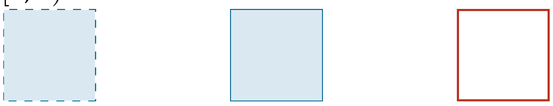
Explanation of Steps: The set Ω includes its bottom and left edges (solid lines) but excludes its top and right edges (dashed lines).

Spoken Protocol [00:05:33 - 00:07:55]

So, how do you draw this example? It would be something like this, right? *[Draws the square on the board]* Is the inside of this square, but you are missing this part of the boundary and this part of the boundary. Now, can you see that these points satisfy this definition here? If I take a point here x here, is on the boundary — so is on the closure, sorry, is on the closure. Because I can take a sequence of points from the inside, from the red part (i.e., the shaded interior), that approaches x, right? But these points that are not in my set because I am excluding them, they are also on the closure. This point here x over here is also on the closure. This belongs to the closure because I can take a sequence from here that approaches my point. Right?

Visualizing Interior, Closure, and Boundary

For the set $\Omega = [0, 1) \times [0, 1)$:



$\Omega^\circ = (0, 1)^2$ $\overline{\Omega} = [0, 1]^2$ $\partial\Omega$

Explanation of Steps: The interior Ω° is the open square $(0, 1) \times (0, 1)$, which excludes all edges. The closure $\overline{\Omega}$ is the closed square $[0, 1] \times [0, 1]$, which includes all edges. The boundary $\partial\Omega$ is the set of all four edges of the square.

Spoken Protocol [00:07:55 - 00:08:52]

So, again all of these are perfectly common sense notions but that we can formalize and make perfectly rigorous just using these few axioms. And then, yes, I will prove maybe I will prove a lemma. *[The lecturer erases the left chalkboard and prepares to write a new lemma]*

11.6 Characterizing Open and Closed Sets via Sequences

Spoken Protocol [00:08:52 - 00:10:12]

This lemma is called open and closed through sequences. How we determine if a set is open and closed using convergent sequences. So... *[Starts writing the lemma on the board]*

Lemma: Open and Closed Sets through Sequences

Lemma 11.48 (Open and Closed Sets through Sequences)**.** *Let (X, d) be a metric space.*

(a) *A subset $U \subseteq X$ is **open** if and only if for every sequence $(x_n)_{n\geq 0}$ in X that converges to a point $x \in U$, the sequence elements x_n belong to U **eventually**.*

$$U \text{ is open} \iff (\forall (x_n) \to x \in U \implies x_n \in U \text{ eventually})$$

(b) *A subset $A \subseteq X$ is **closed** if and only if for every sequence $(x_n)_{n\geq 0}$ contained in A that converges to a point $x \in X$, the limit x also belongs to A.*

$$A \text{ is closed} \iff (\forall (x_n)_{n\geq 0} \subseteq A \text{ s.t. } x_n \to x \implies x \in A)$$

Spoken Protocol [00:10:12 - 00:12:10]

So, this concept of *"up to finitely many exceptions"* is very useful and is very, uh, nice to — to say the thing shortly. And we give a name to this. And you have a more formal definition in the — in the notes. So, when something happens — when you have a sequence and some property, like for instance being in U, happens, is true up to finitely many exceptions, we say that this property happens *"eventually"*.

Definition: Eventually

Definition 11.46 (Eventually)**.** A property $P(x_n)$ is said to hold **eventually** for a sequence $(x_n)_{n\geq 0}$ if it holds for all but finitely many terms. Formally:

$$\exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, P(x_n) \text{ is true}$$

Proof of Lemma 11.48 (Part 1)

Spoken Protocol [00:12:10 - 00:15:50]

Proof of (1). So, I need to do two implications. First I will do this side ($\implies$) that is the one I draw here *[points at the drawing of the open set U]*. Assume that U is open. And now take a point — I'm just going to put in symbols the drawing that I did here. Take x in U, any point. Then there — by definition of open set, since U is open, exists some r positive such that the ball of radius r centered at x is contained in U.
And now, since by assumption x_n is converging to x, by definition of convergence, what was the definition of convergence? Remember: given any ϵ — I called it in the definition — this was the ϵ (i.e., the radius r), right? Given any ϵ, now is r but is the same. For n large enough, bigger than a certain capital N, my sequence will belong to the ball of radius ϵ (i.e., r).

Forward Implication $\implies$

Assume U is open and $x_n \to x \in U$. Since U is open:

$$\exists r > 0 \text{ s.t. } B(x,r) \subseteq U$$

Since $x_n \to x$, by the definition of convergence (using $\epsilon = r$):

$$\exists N \in \mathbb{N} \text{ s.t. } \forall m \geq N, d(x_m, x) < r$$

This is equivalent to saying:

$$\forall m \geq N, x_m \in B(x,r)$$

Since $B(x,r) \subseteq U$, it follows that:

$$\forall m \geq N, x_m \in U$$

Thus, $x_n \in U$ eventually.

Spoken Protocol [00:15:50 - 00:18:16]

Now I will prove the second implication ($\impliedby$). And I will prove this by contraposition. $A \implies B$ is equivalent to $\neg B \implies \neg A$, right? We know that. So, I will prove that *"not left"* (i.e., U is not open) implies *"not right"* (i.e., there exists a sequence $x_n \to x \in U$ such that $x_n \notin U$ infinitely often).
So, now, let's do a bit of logic. What is the negation of U open? Remember now, every time: what is the definition of U open? U open is: for every point in my set U, exists some radius such that the ball is contained in U. Now we do the negation. The negation of *"for every point"* is *"exists a point"*. So, U is not open means exists x in U such that for every radius r positive, the ball centered at x with radius r is not contained in U.

Backward Implication $\Longleftarrow$ (via Contraposition)

We prove that if U is not open, then there exists a sequence $x_n \to x \in U$ such that $x_n \notin U$ for all n.

Negation of Openness:

$$U \text{ is not open} \iff \exists x \in U \text{ s.t. } \forall r > 0, B(x,r) \not\subseteq U$$

The condition $B(x,r) \not\subseteq U$ means that the ball must contain at least one point from the complement $X \setminus U$:

$$\forall r > 0, B(x,r) \cap (X \setminus U) \neq \emptyset$$

Spoken Protocol [00:18:16 - 00:19:15]

[A student asks a question about notation] Ah, you mean this? *[points at the $\subseteq$ symbol on the board]* Is the same. You mean this? *[points at the $\subset$ symbol]* Is the same meaning. So, some people use the equality — since it means the same, I save one line, right? But many — this is very standard, many people just use this. So, I will use always this *"contained"* without the equality sign.

Student Question

You write the subset symbol without the extra line at the bottom. Does it have a meaning, like that it cannot be equal, or is that just notation?

Spoken Protocol [continued]

It's just notation. In this course, $\subset$ and $\subseteq$ are used interchangeably to mean subset.

Spoken Protocol [00:19:15 - 00:21:15]

Okay, so back to the proof. Taking $r = 1/n$ for $n \geq 1$, I produced a sequence of points — so these balls are smaller and smaller — and these points x_n (i.e., the x_r for $r = 1/n$) are in balls closer and closer to x. So, they are converging to the point x. Right? So, I found a sequence that is converging to the point x and that comes from the outside. But then, this is false (i.e., it contradicts the *"right"* side of our implication). Because I'm saying that up to finitely many exceptions, the sequence should be contained in U. And all of the elements of this sequence are outside of U. So, this is false. This contradicts that x_n belongs to U eventually. And it ends the proof of this implication.

Constructing the Counter-Sequence

For each $n \in \{1, 2, 3, \dots\}$, choose a radius $r_n = \frac{1}{n}$. By our assumption, there exists a point:

$$x_n \in B(x, 1/n) \cap (X \setminus U)$$

This construction yields a sequence $(x_n)_{n \geq 1}$ such that:

(a) $x_n \notin U$ for all n.

(b) $d(x_n, x) < \frac{1}{n} \to 0$, so $x_n \to x$.

Since $x \in U$, the *"right"* side of our lemma would require $x_n \in U$ eventually. However, our sequence is entirely outside U, which is a contradiction. Thus, U must be open.

Q.E.D.

Spoken Protocol [00:21:15 - 00:24:04]

So, this proof by contraposition is very useful. Now, part (2) of the lemma is very similar. I leave it as an exercise. Try to do it, again. If you understood well the definitions and you can draw a picture of what's going on, this shouldn't be too difficult. But it can be a bit difficult at the beginning because there are a lot of new definitions. The best is to practice with these exercises.
Let me put another exercise before continuity. Prove that if (X, d) is a complete metric space and A is a closed subset, then A with the same distance restricted is complete. Example: we know that $\mathbb{R}^n$ is a complete metric space. Then any closed subset of $\mathbb{R}^n$ will be a complete metric space when you restrict the distance. And now you have all the ingredients to prove that. It's not too difficult. It's a good exercise.

Exercise: Completeness of Closed Subsets

Exercise

Prove the following proposition:
Proposition 11.47. Let (X, d) be a complete metric space. If $A \subseteq X$ is a closed subset, then the metric subspace $(A, d|_A)$ is also a complete metric space.

Spoken Protocol [00:24:04 - 00:25:30]

Okay, so I see the usual problem: when I see people that are very serious, I don't know if they are completely lost, if they find it super boring, or what is the problem. So, if someone has questions and wants to participate, we can slow down and ask questions. Otherwise, we continue with other stuff. We still need to do continuity, right? Well, let's make the pause. *[The lecturer prepares for the break]*

15-Minute Break

The lecturer announces a break. The video resumes with the lecturer addressing the class after the break.

Spoken Protocol [00:25:30 - 00:26:39]

Okay, I will start if you don't listen... well. So, may you forgive me for the stereotype, but when I studied mathematics in Barcelona, which is a Mediterranean country, people started on time the lecture. So, they were silent. We are in Switzerland, you should be like the champions of starting on time. And I'm seeing the contrary. I'm very surprised. Okay. And you know this is recorded, so this will be the proof of the misbehavior will be there for the history.

Spoken Protocol [00:00:00 - 00:00:08]

...standard, many people just use this [*points at the* $\subseteq$ *symbol on the board*]. So, I will use always this *"contained"* without the equality sign. [*The lecturer begins erasing the chalkboards to prepare for the next section*]

Topic Transition

The lecturer erases the center and right chalkboards. He briefly writes down the exercise mentioned at the end of the previous segment before transitioning to the main topic of the second half of the lecture: Continuity.

Exercise

Proposition 11.49. Let (X, d) be a complete metric space. If $A \subseteq X$ is a closed subset, then the metric subspace $(A, d|_A)$ is also a complete metric space.

Spoken Protocol [00:00:24 - 00:01:25]

Okay, let's do this because, um, maybe you will find more exciting continuous functions. Since this proof over here [*points at the board*] is super similar to the previous one, and is again — you see that these proofs are one-line proofs. It's just using the definitions and manipulating around. So, I give this proof — I left this proof as an exercise. Try to do it, again. If you understood well the definitions and you can draw a picture of what's going on, this shouldn't be too difficult.
But it can be a bit difficult at the beginning because there are a lot of new definitions. You need to learn them. But the best is to practice with these exercises. To try to do them. Because if I do them, um, it looks like a bunch of symbols that all work sometimes, but — but you don't interiorize the definitions. The best — the best way is to try by yourself.

Spoken Protocol [00:01:25 - 00:03:41]

Okay, so I will — I will do something more exciting now, that is continuity. Because you already know... [*The lecturer erases the board and writes the heading "Continuity"*] Okay. Now we start to have the problems with the shadows. Maybe... so, someone, I don't remember who, complained about shadows. Now I can write here, right? Is just when the other blackboard is annoying. Okay. So, now we will see continuity.

11.7 Continuity in Metric Spaces

Spoken Protocol [00:03:41 - 00:05:47]

For continuity, we will define continuous maps. So, we will be in this setup. f will be a map between two sets X and Y. And they are both metric spaces. So, we will have some X with its distance d_X, and Y with its distance d_Y. So, many times for us, these sets will be the Euclidean space of some dimension and the Euclidean space with some other dimension, right? But it can be whatever.
So, this is the setup. Both — both are different metric spaces. Well, different or the same;

they could be the same. Now, definition. We say that a map like this is f is continuous if one of the following three equivalent properties hold true.

Definition: Continuity

Definition 11.52 (Continuity)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A mapping $f : X \to Y$ is said to be **continuous** if it satisfies any of the following three equivalent conditions:

Spoken Protocol [00:05:47 - 00:07:10]

So, now I will state the three definitions of continuity, because at least with two of them you are very familiar from — from Analysis 1, right? But I will state now postulating that they are equivalent, and later we will prove immediately that they are indeed equivalent. Right? So, the definition number one of continuity is the ϵ-δ continuity that you are familiar.
So, you know this very well for functions of one real variable. Now we need to abstract the same notion in metric spaces. So, how was the definition of — of ϵ-δ? So, can someone remind me in the real line what is the definition of continuity with ϵ-δ? Does someone remember? [*The lecturer retrieves the catch-box microphone*] Yeah, good. [*Tosses microphone to student*]

Student Answer

I think it was: a function is continuous at x_0 if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all x, if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$.

Spoken Protocol [continued]

So, up to — is very good. You said what is continuous at a point. For a continuous function is continuous at all the points, you were going to say that. But I will use directly the continuous at — at every point. So, I will start exactly as you said. He said: well, first I will add — since I want to be continuous at every point, I will add *"for every point"*. For all x in X. Okay. Next is what you said. For every x, given — for all ϵ positive, exists δ positive. And now the rest of your definition used intervals and stuff that are not available in metric spaces. So, how we translate?

The Three Faces of Continuity

Proposition 11.53 (The Three Faces of Continuity)**.** Let $f : (X, d_X) \to (Y, d_Y)$ be a map between metric spaces. The following are equivalent:

(a) **ϵ-δ Continuity:**

$$\forall x \in X, \forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x' \in X, d_X(x, x') < \delta \implies d_Y(f(x), f(x')) < \epsilon$$

Spoken Protocol [00:08:23 - 00:10:23]

So, the notion will be that if the distance between x and x' is less than δ, then the distance

between $f(x)$ and $f(x')$ is less than ϵ. And what is this distance? This distance is the distance in the space here, X, and this distance — these guys belong to Y, so this is the distance in Y. Right? What else could be? Cannot be anything else. So, if you think about this, is the only way to generalize.
Another way to say this, because this is a bit annoying to write, is a bit long, is to say that the ball — let me put it like this: this thing, the bracket red, is the same, exactly the same as saying that the f — when I compute f the image of the ball of radius δ centered at x will be contained in the ball of radius ϵ centered at $f(x)$.

Geometric Interpretation of ϵ-δ

The ϵ-δ condition can be expressed concisely using open balls:

$$f(B_\delta(x)) \subseteq B_\epsilon(f(x)) \tag{11.1}$$

Explanation of Steps: This states that the image of a sufficiently small neighborhood around x (the δ-ball) must be entirely contained within a given neighborhood around the image $f(x)$ (the ϵ-ball).

Spoken Protocol [00:10:23 - 00:12:18]

So, this is the ϵ-δ definition. Second definition is with sequences. Right? Continuity with sequences. And — and this is the same as in $\mathbb{R}$ essentially. So, if you have x_n is a sequence and x_n is converging to a point x, then the sequence of the images, that is $f(x_n)$, is convergent in the space Y with limit $f(x)$. So, and a compact way to write this is that $f(x_n)$ this sequence in n converges when n goes to infinity to $f(x)$. And for this essentially you don't even need anything new. So, is really the same definition of $\mathbb{R}$, you plug it here and it makes sense.

Sequential Continuity

(b) Sequential Continuity:

$$\forall (x_n)_{n\geq 0} \subset X, \quad x_n \to x \implies f(x_n) \to f(x)$$

Spoken Protocol [00:12:18 - 00:14:15]

And then there is the new notion of continuity, that is the topological notion of continuity, the number three, that this one is new but will be very easy to prove that is the same. And we will use this all the time, that's why I'm giving. That says that for every V in Y open, the pre-image $f^{-1}(V)$ that is a subset of X is open.
A map, a function is continuous is whenever you take an open set in the target and you see it back — so you go f^{-1} — you get an open set here in the — in the domain. Now, maybe is not obvious that the three are equivalent, but we will show that.

Topological Continuity

(c) Topological Continuity:

$$\forall V \subseteq Y \text{ open} \implies f^{-1}(V) \subseteq X \text{ is open}$$

11.3 Proving the Equivalences

Spoken Protocol [00:14:15 - 00:16:35]

Okay, let's give the proof of this proposition. So, first I need to start to show — so I will start with the ϵ-δ continuity, right? And I will show that (1) implies (2). So, first step I will show that (1) implies (2). By assumption, I am in (1). So, my function f assume that f from X to Y is ϵ-δ continuous. And let x_n be a sequence converging to a point x. And we want to show that $f(x_n)$ converges to $f(x)$. Right?
So, I need to use that is ϵ-δ continuous. So, the first thing is to find the ϵ. So, you give me the ϵ positive. What I want to prove — let me put it like this: I want to prove that the distance between $f(x_n)$ and $f(x)$ is less than ϵ eventually for every ϵ positive. You see, I have rephrased here I have rephrased the notion of convergence. So, what I want to prove really is that $f(x_n)$ converges to $f(x)$. This is my goal.

Proof of Proposition 11.53

Forward Implication $(1) \implies (2)$

Assume f is ϵ-δ continuous. Let $(x_n)_{n\geq 0}$ be a sequence in X such that $x_n \to x$. We want to show that $f(x_n) \to f(x)$, which means:

$$\forall \epsilon > 0, \quad d_Y(f(x_n), f(x)) < \epsilon \text{ eventually}$$

Given $\epsilon > 0$, by the ϵ-δ continuity of f:

$$\exists \delta > 0 \text{ s.t. } f(B_\delta(x)) \subseteq B_\epsilon(f(x))$$

Since $x_n \to x$, by the definition of convergence:

$$x_n \in B_\delta(x) \text{ eventually}$$

Applying the function f to these terms:

$$f(x_n) \in f(B_\delta(x)) \subseteq B_\epsilon(f(x)) \text{ eventually}$$

This implies $d_Y(f(x_n), f(x)) < \epsilon$ eventually, which proves that $f(x_n) \to f(x)$.

Spoken Protocol [00:16:35 - 00:18:12]

But I can rephrase this goal like this: I want to prove that the distance between — sorry, last ten minutes start to... so this is with f. Now is correct. So, what I want to prove is that $f(x_n)$ converges to $f(x)$. This is my goal. But I am putting it in this way because here at least I see the ϵ. So, given the ϵ I can find the δ. So, now, given ϵ positive, by ϵ-δ continuity exists some δ positive such that the ball of radius δ centered at x when I apply f to this ball will be contained in the ball of radius ϵ centered at $f(x)$.

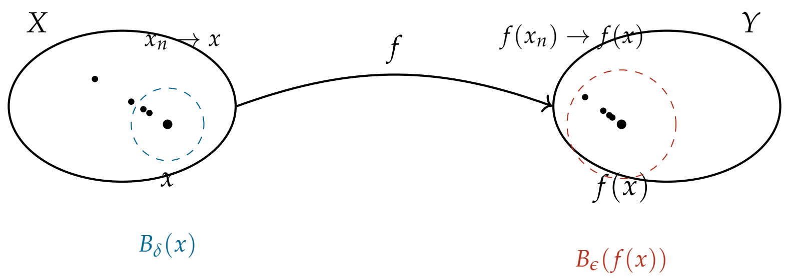
Spoken Protocol [00:18:12 - 00:19:50]

But remember my assumption: my assumption is that x_n was converging to x. So, since δ is positive, x_n belongs to this set (i.e., $B_\delta(x)$) eventually. When I say eventually is like for n very large, x_n will belong to this set. But then, when I read here, the image then this implies that $f(x_n)$ belongs to the ball of radius ϵ centered at $f(x)$ eventually. So, for n large. And this is what I wanted to show. Right? So, this proof shows that (1) implies (2).

Spoken Protocol [00:19:50 - 00:21:15]

So, since this is maybe very symbolic, let me do a picture of what is going on. So, the pictures are never rigorous but they help a lot to — to do some intuition. And even if I don't finish the proof today of all the equivalences, is good to go step by step. So, some people work very good with these proofs that are symbols. For some people like myself, for me are very difficult to think directly with these symbols. So, when I write this kind of proof before writing to the notes or something, what I do is I do the drawing. First I do the drawing and then I translate the drawing in the symbols. Directly the symbols for some people is easy, for some people is more difficult. So, it depends on which way you reason.

Visualizing Continuity



Geometric Interpretation of Continuity: This diagram clearly illustrates the ϵ-δ definition of continuity in metric spaces. The sequence (x_n) in X converges to x, and its image $(f(x_n))$ in Y converges to $f(x)$. The dashed circles represent open balls, showing that for any ϵ-ball around $f(x)$, there exists a δ-ball around x whose image is entirely contained within the ϵ-ball.

Spoken Protocol [00:22:32 - 00:24:26]

So, what is going on in this proof? We have our first metric space X and our second metric space Y. And we have a continuous map from X to Y. And then what we wanted to show is that whenever we have a sequence here approaching x, the corresponding sequence of images will approach the $f(x)$. And we had to reason with the ϵ-δ.
So, what this that this sequence converges here means literally that whenever I pick any ball — can be very, very tiny this ball, very small — then if I wait enough, all the elements of the sequence will fall inside of my ball. Right? If the ball is smaller, I need to wait more and more. But it always falls. And now, this is what I want to prove. So, to find this ball and to find that if I wait long enough I will be here, I need to use that the x_n is convergent to x. So, I need to go back with this ball and see what happens with this ball, right? So, what happens is that this ball here comes from some set here around x that may not be a ball. But what I know is that here there is a ball of radius δ. This is the ϵ-δ. The blue ball. That when I compute the image of this ball, maybe it becomes a *"potato"* or something, but is inside of my original ball. Right? And since this is a ball, the yellow points eventually will be in this ball, and then when I take the image, the points will be in my ball because the blue is inside of the bigger ball. So, with drawings is a bit clearer for some people perhaps.

Spoken Protocol [00:24:26 - 00:24:53]

[*A student asks a question*] Ah, there is some... [*The lecturer retrieves the catch-box microphone*]

Student Question

Can I give back the cube eventually?

Spoken Protocol [continued]

[*The lecturer laughs*] Ah, you need to get — this was the question, to get back the cube? No. So, or someone else has a question. Who has a question? No, I had the cube the whole time and I was waiting for you to... ah, okay, okay.

Spoken Protocol [00:24:53 - 00:26:13]

Okay, now let's start to prove that (2) implies (3). To prove that (2) implies (3), we will actually prove that *"not 3"* implies *"not 2"*. So, again contraposition. Remember: what was 3? 3 was saying that for every V subset Y open, $f^{-1}(V)$ subset X is open. Now, what is *"not 3"*? The negation of this. The negation of *"for all"* is *"exists"*. So, *"not 3"* means exists some V subset Y open such that $f^{-1}(V)$ is not open. Right?

Forward Implication (2) $\implies$ (3) (via Contraposition)

We prove $\neg(3) \implies \neg(2)$. Assume (3) is false. Then there exists an open set $V \subseteq Y$ such that its pre-image $U = f^{-1}(V)$ is **not open** in X.

Spoken Protocol [00:26:13 - 00:28:12]

So, what does it mean not open? Not open, we said before. Not open, it means that exists some point x in $f^{-1}(V)$ such that and there is a sequence x_n convergent to x — this is what I did before, remember when I said a set is not open, there are a sequence of balls that have some point in the complement, right? So, there is a sequence x_n convergent to x and all the points x_n do not belong or belong to the complement. Right? So, this gives me a sequence. And now, I will try to use this sequence to contradict (2).

Constructing the Counter-Sequence

Since $U = f^{-1}(V)$ is not open, by Lemma 11.48:

$$\exists x \in U \text{ and a sequence } (x_n)_{n\geq 0} \subset X \setminus U \text{ s.t. } x_n \to x$$

By the definition of the pre-image:
- $x \in f^{-1}(V) \implies f(x) \in V$.
- $x_n \notin f^{-1}(V) \implies f(x_n) \notin V$ for all n.

Spoken Protocol [00:28:12 - 00:29:34]

So, this sequence is converging to x, so I need to try to show that the images do not converge to $f(x)$. But then $f(x_n)$ belongs all belong to the complement of the image (i.e., actually the complement of V). And you notice that V because x belongs to $f^{-1}(V)$, $f(x)$ belongs to V. $f(x)$ belongs to an open set. And what I said before about open sets? That a set is open if and only if whenever you have a sequence eventually belongs to your set. But here all the sequence is in the complement of the set. So, $f(x_n)$ cannot converge to $f(x)$. Right? I use the characterization of open set by sequences. This point is inside my open set and the points are outside. So, these points can never converge to this one. Right? So, this gives that the negation of (2).

Contradicting Sequential Convergence

We have $f(x) \in V$ and $f(x_n) \in Y \setminus V$ for all n. Since V is open, by Lemma 11.48, any sequence converging to $f(x)$ must eventually be contained in V. However, our sequence $(f(x_n))$ is entirely contained in the complement $Y \setminus V$. Therefore, $f(x_n) \not\to f(x)$. This contradicts condition (2), completing the proof that (2) $\implies$ (3).

Q.E.D.

Spoken Protocol [00:29:34 - 00:29:59]

And since it is almost time, actually this is the easiest of the three, but the next day we will show that (3) implies (1) and we will close the — we will close the circle. Okay? And since it is maybe the easiest, you can try to do as an exercise. If you succeed, less work for me. *[The students applaud as the lecturer packs up his equipment]*